

PROJECT EVALUATION

MEAL PLAN • RECOMMENDATION SYSTEM



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PITCH DECK PRESENTATION



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PROBLEM DESCRIPTION

INTRODUCTION TO THE PROBLEM

Maintaining a healthy diet in Sri Lanka is challenging due to cultural food preferences, varying income levels, and unique dietary needs. Many struggle to find affordable, nutritious meal plans and personalized nutrition guidance that considers health concerns, allergies, fitness goals, and budget constraints is often lacking.

IMPACT OF THE PROBLEM

- **Health Risks:** Poor diet management leads to obesity, diabetes, and heart disease in Sri Lanka.
- **Limited Awareness:** A lack of personalized nutrition advice worsens the problem.
- **Imbalanced Diets:** Without tailored meal plans, people are more likely to eat unhealthily.
- **Long-term Issues:** This can result in chronic health problems.



A photograph of two men sitting at a desk. The man on the left, wearing a black leather jacket, is holding a smartphone. The man on the right, wearing a white jacket, is writing in a notebook with a pen. There is a coffee cup and some papers on the desk. The background is slightly blurred.

SOLUTION

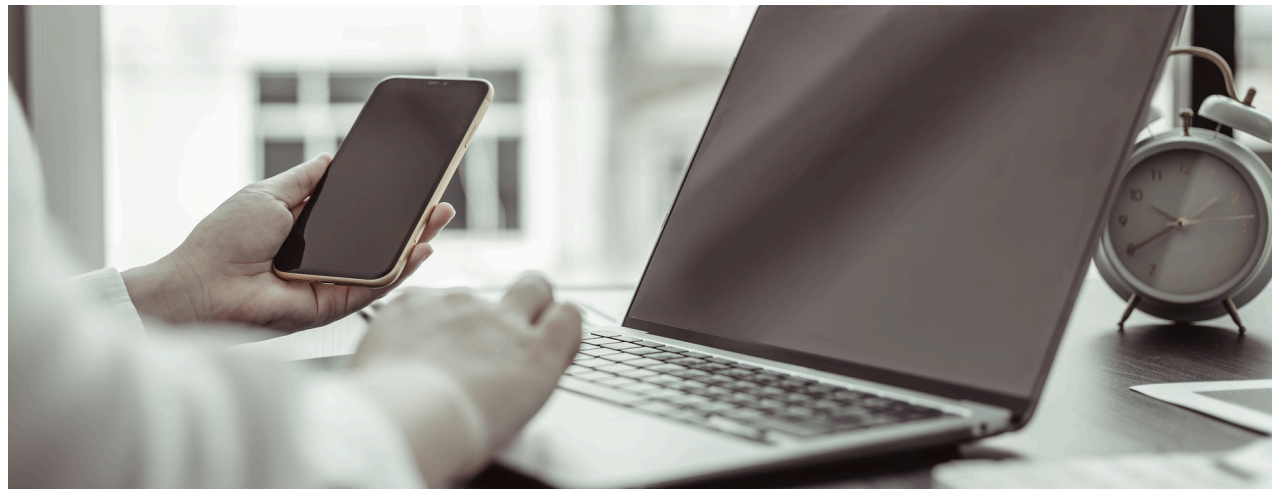
PERSONALIZED MEAL PLAN

The Cool Meal App delivers personalized AI-driven meal plans based on user health, fitness, and budget. Users get real-time suggestions via chatbot, while admins manage plans through an admin panel.

TECHNOLOGY

Leveraging AI, the app includes an intelligent chatbot for meal suggestions and a seamless mobile interface for users. The admin panel allows backend customization of meal plans and user data monitoring, providing both users and administrators with powerful, flexible tools.

OBJECTIVE OF THE PROJECT



01

Project Focus

Develop a meal plan recommendation system tailored for the Sri Lankan population.

02

Personalized Meal Plans

Generate meal plans based on each user's health concerns, dietary restrictions, fitness goals, and budget.

03

Enhanced Chatbot

Integrate a chatbot to provide personalized nutrition advice and support.

04

Admin Panel

Implement an admin panel to manage user data, customize meal plans, Add new data to chatbot and monitor system performance.

WHAT WE HAVE BUILT?

01

Meal Plans

Machine learning tailors personalized meal plans.

02

Mobile app

User-friendly mobile app for seamless access.

03

Admin Panel

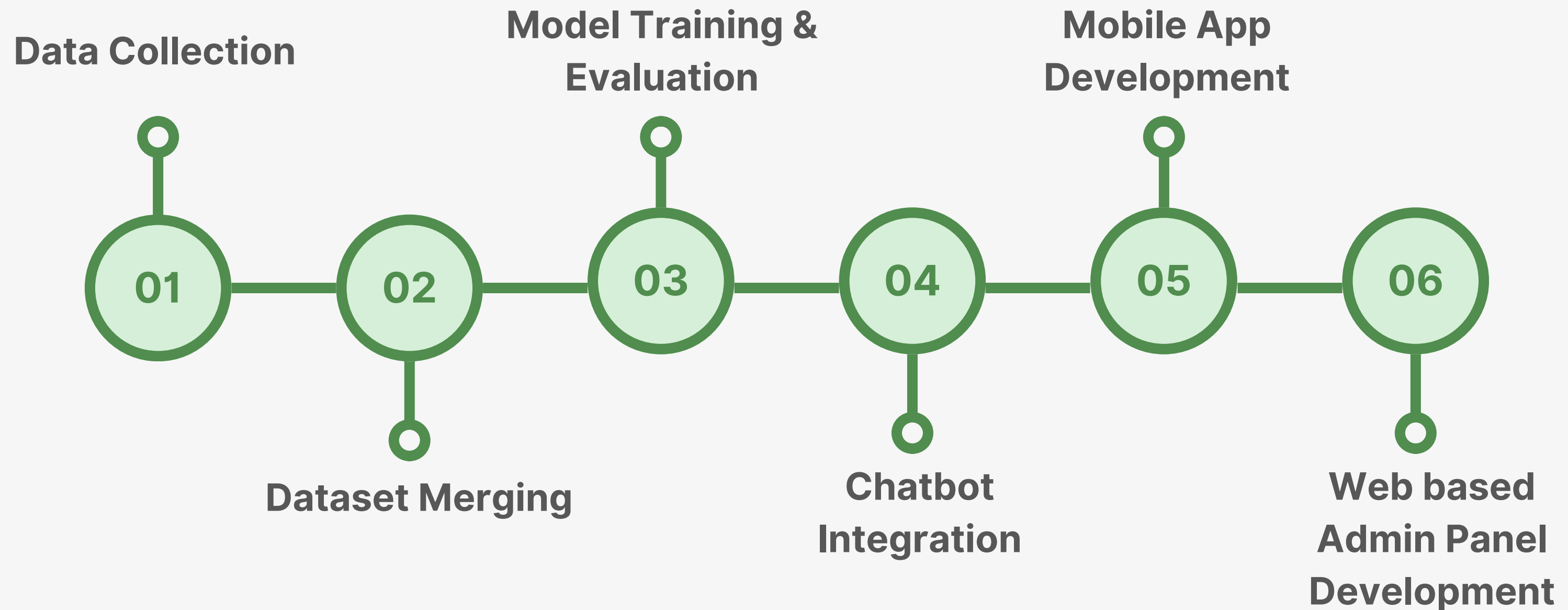
Manage and customize meal plans easily.

04

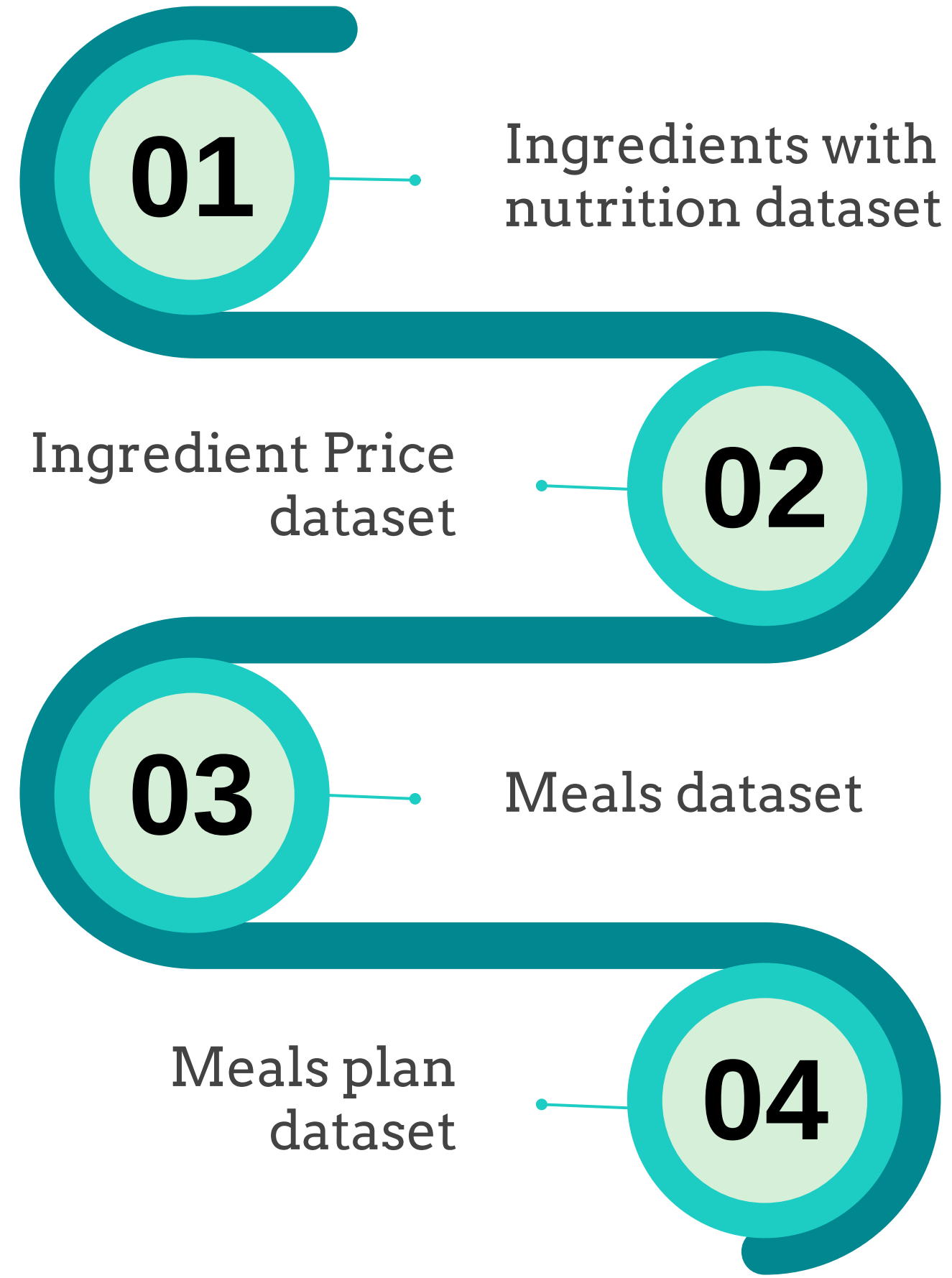
ChatBot

AI-powered chatbot provides real-time suggestions.

DEVELOPMENT STEPS



DATA SOURCES AND DATA PREPARATION



DATA SOURCES AND DATA PREPARATION

INGREDIENTS WITH NUTRITIONS

Sheet 1												
	A	B	C	D	E	F	G	H	I	J	K	L
1	Food_Code	Price_Code	Food_Name	Energy(Kcal)	Protein(g)	Total fat(g)	Carbohydrates(g)	Total Dietary	Vitamin A(µg)	Vitamin D(µg)	Vitamin K(µg)	Vitamin E(r
2	SLA001	P1069	Barley (Hordeum vulgare)	313	11.4	1.18	60.9	15.48	0	0	1.98	0
3	SLA005	P1072	Kurakkan	322	7.11	1.77	67.45	11.16	0.22	41.82	3.25	0
4	SLA006	P1071	habalapethi	350	7.29	1.17	76.08	3.93	0	0	0.95	0
5	SLA007	P1070	Rice, kaluhinati (Oryza sativa)	348	9.88	1.9	72.28	5.21	0	0	2.75	0
6	SLA008	P0624	Rice, keeri samba (Oryza sativa)	368	8.47	1.84	79.41	1.17	0	0	2.56	0
7	SLA009	P0054	Rice, kekulu, red (Oryza sativa)	366	8.64	2.12	77.59	3.18	0	0	2.67	0
8	SLA010	P0115	Rice, kekulu, white (Oryza sativa)	361	9.88	1.52	76.67	2.36	0	0	2.85	0
9	SLA011	P1073	Rice, nadu, white (Oryza sativa)	366	7.76	1.48	80.11	2.45	0	0	2.91	0
10	SLA012	P0617	Rice, samba (Oryza sativa)	369	6.86	2.45	79.53	3.24	0	0	2.23	0
11	SLA013	P0617	Rice, samba kekulu (Oryza sativa)	355	8.81	1.53	76.12	4.42	0	0	2.52	0
12	SLA014	P0629	halpiti noodles	365	7.29	1.13	80.97	2.68	0	0	1.64	0
13	SLA015	P0629	Noodles	332	9.31	0.42	70.69	9.81	0.33	4.19	1.35	0
14	SLA018	P1133	Wheat, flour, refined (Triticum aestiv	355	10.42	0.73	75.25	2.99	0.49	5.67	0.95	0
15	SLA019	P1134	Wheat, semolina (rulang)	332	10.85	0.72	68.83	9.57	0.24	7.09	1.4	
16	SLH001	P0610	Egg	170	13.14	13	0	0	208	4.46	64	
17	SLH006	P0611	Chicken	176	23.14	9.24	0	0	6.88	0	31.65	0
18	SLH016	P1128	Beef	160	23.3	7.45	0	0	2.15	0	7.8	
19	SLH019	P1129	Pork, shoulder	161	18.83	9.5	0	0	1.9	5.63	3.02	0
20	SLF001	P1082	Anchovy (Stolephorus indicus)	87	19.88	0.78	0	0	12.18	3.31	0.6	0
21	SLF003	P1114	Barracuda, Pickhandle (Sphyraena j	107	22.46	1.74	0	0	24.82	2.27	1.72	0
22	SLF010	P1101	Indian mackerel (Rastrelliger kanagu	101	21.51	1.53	0	0	16.34	1.37	5.25	0
23	SLF013	P1106	Marlin (Istiompax indica)	97	22.89	0.56	0	0	21.51	1.28	0.89	0
24	SLF014	P1093	Mauve-lip thread fin bream (Nemipt	126	22.07	4.2	0	0	25.61	1.84	1.02	0

BASED ON RESEARCH BOOKS FOUND ONLINE

DATA SOURCES AND DATA PREPARATION

INGREDIENTS WITH PRICES

Sheet 1				
	A	B	C	D
1	Price_ID	Title	Unit	Price_Per_100_Unit
2	P0001	Marina Cooking Oil 500ml	ml	137
3	P0002	Pure Dale Kirithe 40g	g	220
4	P0003	Munchee Baby Rusks 60g	g	266.67
5	P0004	Munchee Crunchy Carols 300g	g	133.33
6	P0005	Fadna Shape Up Tea 15 Bags		840
7	P0006	Fadna Tummy Tea 15 Bags		840
8	P0007	Eden Foods Organic Sesame Oil		1900
9	P0008	Garlic	g	69.6
10	P0009	Kist Apple Nectar 200ml	ml	40
11	P0010	Kist Wood Apple Nectar 200ml	ml	60
12	P0011	Kist Aloe Vera Drink 500ml	ml	50
13	P0012	Happy Cow Cheese Wedges 120g	g	613.33
14	P0013	Kotmale Cheese Processed Slices 1Kg	g	420
15	P0014	MD Vanilla Pudding 150g	g	100
16	P0015	MD Mustard Cream 360g	g	202.78
17	P0016	MD Mango Chutney 460g	g	86.96
18	P0017	Magic Ice Cream Fruit & Nut 1L	ml	65
19	P0018	Magic Ice cream Vanilla & Chocolate 1L	ml	48
20	P0019	Magic Ice cream Chocolate 1L	ml	47
21	P0020	Magic Ice cream Vanilla 1L	ml	48
22	P0021	Magic Ice cream Chocolate 2L	ml	49
23	P0022	Magic Ice cream Vanilla & Chocolate 2L	ml	42.5
24	P0023	Magic Ice Cream Fruit & Nut 2L	ml	57.5
25	P0024	Magic Ice cream Vanilla 2L	ml	40
26	P0025	Ruhunu Goraka Pieces 100g	g	120
27	P0029	MDK Vegetable Chinese Rolls (25 Rolls) 1250g	g	100

**BASED ON KEELS DATABASE
&
WEB SCRAPED DATA**

- **SELINAWAMUCII.COM**
- **SALESCOM.LK**

DATA SOURCES AND DATA PREPARATION

MEAL ITEMS

+ Sheet 1											
		B	C	D	E	F	G	H	I	J	
1	al_Id	Name	Meal_Ingredient_Ids	Meal_Ingredients_Names	Meal_Ingredient_Quantities	Category	Description	Supported_Main_Meals	Combir	Morning_Probability	Lunch
2	1	Kottu Roti	SLI019,SLB002,SLD055,SLH006,SL	Roti,carrot,leeks, Chicken,Coconut Oil,	30,20,27.5,20.5,2.5	Main	Sri Lankan street food		1		1
3	2	Hoppers	SLI019,SLI005,SLI011,SLK004	Rice flour, Coconut milk, Sugar,Coconu	70, 10, 5,15,	Main	Delicate rice flour pancakes		0		8
4	3	Kiribath	SLA009,SLI005	Rice, Coconut milk	85,15	Main	Traditional coconut milk rice		1		9
5	4	Fish Ambul Thiyal	SLF025,SLK005,SLK006	Fish, Goraka, Spices	80,10,10	Side	Sour fish curry	3, 60, 61, 62, 151, 171	0		
6	5	Dhal Curry	SLC009,SLI005,SLK006,SLK019	Dhal, Coconut milk, Spices,Termeric po	80,10,5,5	Side	Spicy lentil curry	3, 60, 61, 62, 151, 171	0		
7	7	Pol Sambol	SLI005,SLK002,SLD060,SLE028,SL	Coconut, Chili, Onion,Lime,Garlic	50,5,10,3,3	Side	Coconut and chili relish	12, 60, 171	0		
8	8	Pittu	SLI019,SLI005	Rice flour, Coconut	25,10	Main	Steamed coconut rice flour		1		8
9	9	Lamprais	SLA008,SLH006,SLK006	Rice, Meat, Spices	100,30,10	Main	Dutch-influenced rice and curry		1		4
10	10	Parippu	SLC009,SLI005,SLK019	Dhal, Coconut milk,Termeric powder	10,10,5	Side	Creamy spiced lentil dish	12, 25, 60, 151, 171	0		
11	11	Wambatu Moju	SLD019,SLI017,SLD060,,SLD082	Eggplant, Vinegar,onion,tomato sause	20,0.5,5,5	Side	Sweet and tangy pickle	25, 26, 27	0		
12	12	String Hoppers	SLI019	Rice flour	85	Main	Rice noodle bundles		1		4
13	13	Jackfruit Curry	SLD049,SLI005,SLK009	Jackfruit, Coconut milk,curry leaves	15,10,5	Side	Tender jackfruit in spices	12, 60	0		
14	14	Gotu Kola Sambol	SLD048,SLI005,SLD060	Gotu Kola, Coconut, Onion	10,15,15	Side	Herbal green salad	60	0		
15	15	Eggplant Curry	SLD019,SLK006,SLI005,	Eggplant, Spices, Coconut milk	30,2,10	Side	Spicy eggplant dish	60	0		
16	16	Lunu Miris	SLD060,SLK002,SLE028,SLK011	Onion, Chili, Salt,Lime,Garlic	20, 50, 30	Side	Spicy onion relish	3, 8, 61, 62, 151	0		
17	17	Seeni Sambol	SLD060,SLI011,SLK002	Onion, Sugar, chillii	50,5,5	Side	Sweet caramelized onions	3, 151	0		
18	18	Beetroot Curry	SLB001,SLI005,SLK006,SLK009	Beetroot, Coconut milk, Spices,curry le	30,20,5,2	Side	Sweet and spicy beetroot	60	0		
19	19	Fish Curry	SLF013,SLI005,SLK006,SLK009	Fish, Coconut milk, Spices,curry leaves	50,20,3,2	Side	Spicy coconut fish curry	60, 61, 62, 151, 171	0		
20	21	Chicken Curry	SLH006, SLI005,SLK006,SLK009	Chicken, Coconut milk, Spices,curry le	50, 20,5,3	Side	Spicy chicken curry	1, 3, 9, 12, 25, 26, 27, 28, 31, 32	0		
21	22	Bathala Curry	SLB010,SLI005,SLK006,SLK009	Sweet potato, Coconut milk, Spices,cu	50, 20,5,3	Side	Sweet potato curry	60, 171	0		
22	23	Malu Mirisata	SLF013,SLK002,SLK009	Fish, Chili, curry leaves	50, 10, 3	Side	Chili fish curry	60, 61, 62, 151, 152, 171	0		
23	24	Elumas Curry	SLH018,SLI005,SLK006,SLK009	Goat, Coconut milk, Spices,curry leave	50,20,5,2	Side	Goat meat curry	60, 151, 152, 171	0		
24	25	Vegetable Fried Rice	SLA008,SLB002,SLD055,SLD060,S	Rice,carrot,leeks,onion, Spices,curry le	100,20,20,5,3,2.5	Main	Mixed vegetable rice		1		4
25	26	Chicken Fried Rice	SLA008,SLH006,SLD060,SLB002,S	Rice, Chicken, onion, carrot,leeks,Spic	100,50,20,20,5,5	Main	Spicy chicken rice		1		4
26	27	Egg Fried Rice	SLA008,SLH001,SLD060,SLK006,S	Rice, Egg, onion, Spices,curry leaves,c	100,30,10,5,3,2.5,20,20	Main	Egg mixed fried rice		1		4
27	28	Seafood Fried Rice	SLA008,SLI015,SLF045,SLF025,,SL	Rice,shrimp,crab, Oninon,Spices,curry	100,10,10,5,3,2	Main	Mixed seafood rice		1		4
28	29	Deville Chicken	SLH006,SLK002,SLK009,SLD060	Chicken, Chili, Curry leaves,Onion	50,5,10,20	Side	Sweet and spicy chicken	1, 25, 26, 27, 28, 31, 32, 33, 34,	0		
29	30	Deville Fish	SLF025,SLK002,SLK009,SLD060	Fish, Chili, Curry leaves,Onion	50,5,5,20	Side	Sweet and spicy fish	1, 25, 26, 27, 28, 31, 32, 33, 34,	0		
30	31	Vegetable Kottu	SLA018, SLB002, SLD055, SLK002	Flour, Carrot, leeks, Chilli	70, 20, 20, 20	Main	Mixed vegetable stir-fry		1		1
31	32	Chicken Kottu	SLA018, SLB002, SLD055, SLH006	Flour, Carrot, leeks, Chicken, Chilli	70, 20, 20, 40, 20	Main	Spicy chicken stir-fry		1		1
32	33	Fish Kottu	SLA018, SLB002, SLD055, SLF019,	Flour, Carrot, leeks, Fish, Chilli	70, 20, 20, 40, 20	Main	Spicy fish stir-fry		1		1
33	34	Egg Kottu	SLA018, SLB002, SLD055, SLH001,	Flour, Carrot, leeks, Egg, Chilli	70, 20, 20, 40, 20	Main	Spicy egg stir-fry		1		1
34	35	Seafood Kottu	SLA018, SLB002, SLD055, SLF019,	Flour, Carrot, leeks, Fish, Prawn,Chilli	70, 20, 20, 30,30, 20	Main	Mixed seafood stir-fry		1		1
35	36	Chicken Biryani	SLA008,SLH006,SLD060,SLK006,S	Rice, Chicken, onion, Spices,curry leav	100,50,20,5,3	Main	Spiced chicken rice		1		1

- BASED ON RECIEPEE BOOKS
- CALCULATE PRICE & NUTRITIONS ALGORITHMICALLY

DATASET MERGING

MEALS

	A	B	C	D	E	F	G	H
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	Main Meal	Side Meal	Breakfast_Probability	Lunch_	Dinner_	Combi	Complete Meal	Ingredients
2	Kottu Roti	Chicken Curry	1.0	2.5	8.0	1	Kottu Roti with Chicken Cu	Roti, carrot, leeks, Chicken, Coconut Oil, Chilli Powder, Coc
3	Kottu Roti	Devilled Chicken	1.0	2.5	8.0	1	Kottu Roti with Devilled Chi	Roti, carrot, leeks, Chicken, Coconut Oil, Chilli Powder, Curr
4	Kottu Roti	Devilled Fish	1.0	2.5	8.0	1	Kottu Roti with Devilled Fis	Roti, carrot, leeks, Chicken, Coconut Oil, Chilli Powder, Fish,
5	Kottu Roti	Seer Fish Curry	1.0	2.5	8.0	1	Kottu Roti with Seer Fish C	Roti, carrot, leeks, Chicken, Coconut Oil, Chilli Powder, Seer
6	Kiribath	Seeni Sambol	9.0	1.0	0.0	1	Kiribath with Seeni Sambol	Rice, Coconut milk, Onion, Sugar, chillii
7	Kiribath	Chicken Curry	9.0	1.0	0.0	1	Kiribath with Chicken Curry	Rice, Coconut milk, Chicken, Coconut milk, Spices, curry le
8	Kiribath	Fish Ambul Thiyal	9.0	1.0	0.0	1	Kiribath with Fish Ambul Th	Rice, Coconut milk, Fish, Goraka, Spices
9	Kiribath	Dhal Curry	9.0	1.0	0.0	1	Kiribath with Dhal Curry	Rice, Coconut milk, Dhal, Spices, Termeric powder
10	Kiribath	Lunu Miris	9.0	1.0	0.0	1	Kiribath with Lunu Miris	Rice, Coconut milk, Onion, Chili, Salt, Lime
11	Pittu	Lunu Miris	8.0	6.0	4.0	1	Pittu with Lunu Miris	Rice flour, Coconut, Onion, Chili, Salt, Lime
12	Lamprais	Chicken Curry	4.0	9.0	9.0	1	Lamprais with Chicken Cur	Rice, Meat, Spices, Coconut milk, curry leaves
13	String Hoppers	Spicy Crab	4.0	4.0	4.0	1	String Hoppers with Spicy C	Rice flour, Crab, Oil, chillies
14	String Hoppers	Parippu	4.0	4.0	4.0	1	String Hoppers with Paripp	Rice flour, Dhal, Coconut milk, Termeric powder
15	String Hoppers	Thalapath Fish Curry	4.0	4.0	4.0	1	String Hoppers with Thalap	Rice flour, Thalapath fish, Coconut milk, Spices
16	String Hoppers	Prawn Curry	4.0	4.0	4.0	1	String Hoppers with Prawn	Rice flour, Prawn, Coconut milk, Chillies
17	String Hoppers	Cuttlefish Curry	4.0	4.0	4.0	1	String Hoppers with Cuttlef	Rice flour, Cuttlefish, Coconut milk, Spices
18	String Hoppers	Kiri Hodi	4.0	4.0	4.0	1	String Hoppers with Kiri Ho	Rice flour, Coconut milk
19	String Hoppers	Potato Curry	4.0	4.0	4.0	1	String Hoppers with Potato	Rice flour, Potato, Coconut milk, Spices, curry leaves, Turme
20	String Hoppers	Egg Curry	4.0	4.0	4.0	1	String Hoppers with Egg C	Rice flour, Egg, Coconut milk, Chillies
21	String Hoppers	Kochchi Sambol	4.0	4.0	4.0	1	String Hoppers with Kochch	Rice flour, Green chili, Coconut
22	String Hoppers	Pol Sambol	4.0	4.0	4.0	1	String Hoppers with Pol Sa	Rice flour, Coconut, Chili, Onion, Lime, Garlic
23	String Hoppers	Jackfruit Curry	4.0	4.0	4.0	1	String Hoppers with Jackfru	Rice flour, Jackfruit, Coconut milk, curry leaves
24	String Hoppers	Coconut Sambol	4.0	4.0	4.0	1	String Hoppers with Cocon	Rice flour, Coconut, Chili, Onion
25	String Hoppers	Chicken Curry	4.0	4.0	4.0	1	String Hoppers with Chicke	Rice flour, Chicken, Coconut milk, Spices, curry leaves

- BASED ON MEAL BOOKS
- CALCULATE PRICE & NUTRITIONS ALGORITHMICALLY

DATASET
MERGING

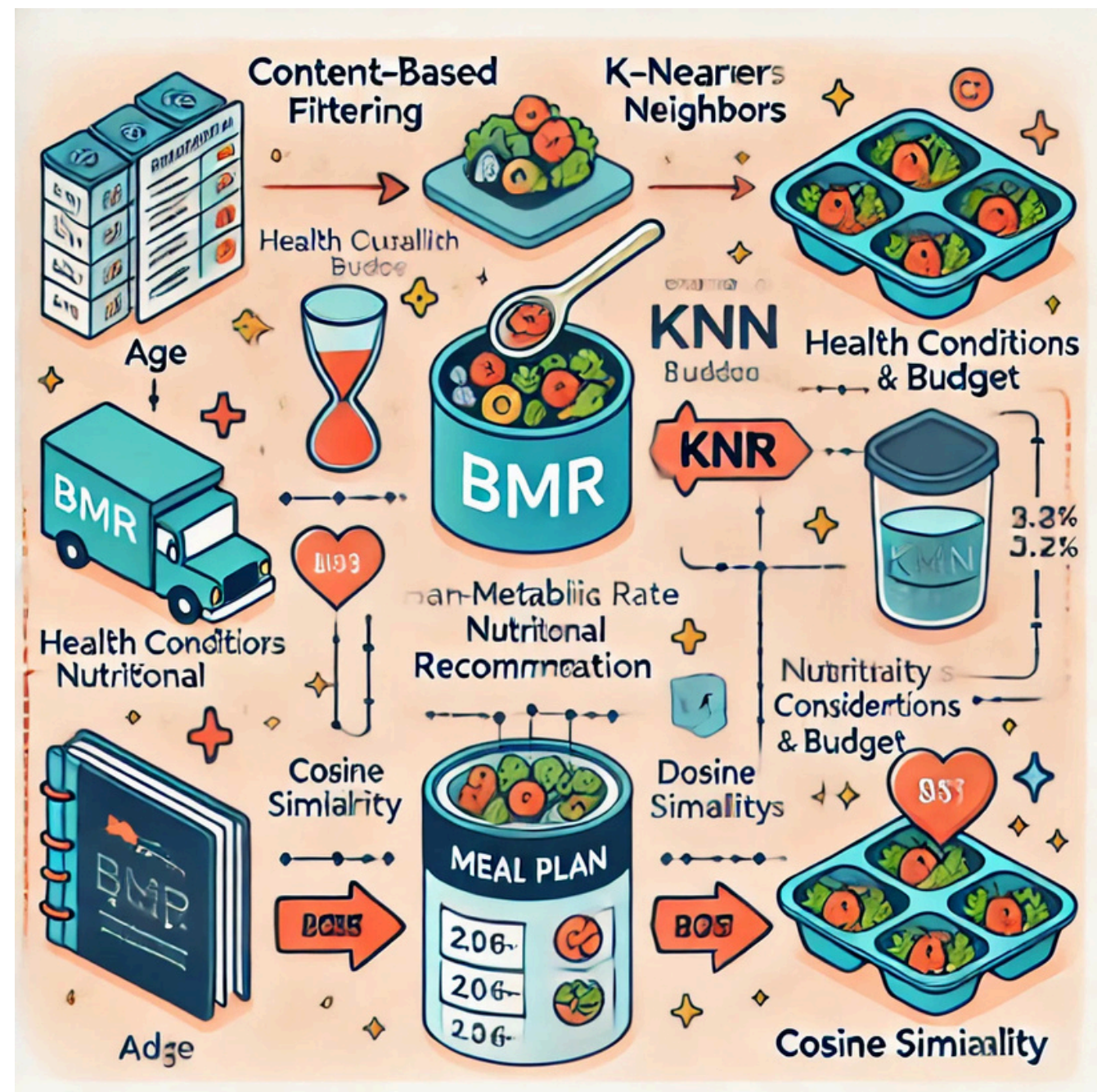
MEAL PLANS

Sheet 1						
Breakfast	Lunch	Dinner	Price	Total Energy(Kcal)	Total Protein(g)	Total fat(g)
Red Rice with Fish Curry, Tapioca Chips, Batu Moju	Red Rice with Banana Blossom Curry, Green Gram Curry, Fenugreek Curry	Vegetable Kottu with Devilled Fish	792.3	1482.37	73.5357	83.1561
Red Rice with Parippu, Brinjal Moju, Ginger Chicken	Kottu Roti with Chicken Curry	Red Rice with Fish Curry, Kiri Hodi, Sprats Fry	748.878571428	1403.335	72.6759	116.7930
Red Rice with Coriander Sambol, Coconut Sambol, Shrimp Cutlets	Red Rice with Chickpea Curry, Fenugreek Curry, Egg Curry	Vegetable Biryani with Chicken Curry, Devilled Fish	755.2	2006.6580000000000	95.27754	154.7375
Hoppers	Red Rice with Elumas Curry, Potato Curry, Devilled Beef	Pumpkin Soup	408.125	1273.78	39.334	111.29760000000000
Red Rice with Bathala Curry, Mustard Greens Curry, Murukku	String Hoppers with Pol Sambol	Coconut Roti with Dhal Curry	304.678	1378.88	57.9928	93.033
Red Rice with Egg Curry, Radish Sambol, Polos Mallum	Gotu Kola Kanda	Red Rice with Green Pea Curry, Vegetable Cutlets, Polos Mallum	275.59	1271.5500000000000	44.792500000000000	121.19
Seafood Fried Rice with Chicken Curry, Devilled Fish	Red Rice with Snake Gourd Curry, Pumpkin curry, Jaffna Crab Curry	Red Rice with Chicken Curry, Fenugreek Curry, Jaffna Crab Curry	650.347	1560.6800000000000	83.2483	124.253
Red Rice with Eggplant Curry, Spinach Curry, Radish Sambol	Red Rice with Devilled Fish, Carrot Sambol, Kiri Hodi	Red Rice with Dhal Curry, Collard Greens Curry, Jackfruit Mallum	2975.68	1538.3400000000000	66.96960000000000	149.736
Red Rice with Gotu Kola Sambol, Mint Sambol, Seeni Sambol	Red Rice with Chicken Curry, Bean Curry, Batu Moju	Seafood Biryani with Ginger Chicken	602.388	1558.3320000000000	75.53611000000000	87.84216000000000
Red Rice with Snake Gourd Curry, Green Pea Curry, Pumpkin Leaves Curry	Vegetable Kottu with Devilled Chicken	Zesty Lime Rice	373.465	1237.9000000000000	50.801500000000000	58.825
Red Rice with Bean Curry, Thalapath Fish Curry, Simmered Fish Roe	Pol Roti with Thalapath Fish Curry	Red Rice with Pineapple Curry, Pineapple Pickle, Polos Ambula	1018.55	1652.1500000000000	101.80850000000000	134.315
Red Rice with Lotus Root Curry, Vegetable Balls, Pumpkin curry	Red Rice with Dhal Curry, Green Pea Curry, Leeks Curry	Red Rice with Prawn Curry, Carrot Sambol, Garlic curry	483.0	1522.5	69.90150000000000	108.41850000000000
Red Rice with Tropical Salad, Polos Ambula, Indian Fish Curry	Vegetable Kottu with Chicken Curry	Egg Hoppers	385.731	1163.04	56.8977	80.50870000000000
Red Rice with Potato Curry, Gotu Kola Sambol, Garlic curry	Red Rice with Banana Blossom Curry, Coriander Sambol, Pineapple Pickle	Red Rice with Soya Meat Curry, Leeks Curry, Pumpkin Leaves Curry	518.21800000000000	1524.7200000000000	48.3652	149.218
Chicken Koththu with Ginger Chicken	Red Rice with Ash Plantain Curry, Vegetable Cutlets, Spicy Crab	Prawn Tempered	517.79	1275.85	53.369	79.50600000000000
Kiribath with Seeni Sambol	Prawn Tempered	Fish Biryani with Chicken Curry, Ginger Chicken	731.305	1338.13	69.7864	60.374
Red Rice with Green Gram Curry, Lotus Root Curry, Shrimp Cutlets	Red Rice with Fish Curry, Beetroot curry, Polos Mallum	Chicken Fried Rice with Devilled Chicken, Vegetable Balls	784.385	1753.77	91.0722	105.728
Lavariya	Egg Fried Rice with Cashew Curry, potatoe cheese Balls	Seafood Fried Rice with Ginger Chicken, Tapioca Chips	958.8560000000000	2032.1400000000000	65.44885000000000	101.3649
Red Rice with Malu Mirisata, Vegetable Balls, Murukku	Vegetable Biryani with Devilled Fish, Ginger Chicken	Fish Kottu with Ginger Chicken	1067.1700000000000	1273.508	87.17754000000000	20.6205
Coconut Roti with Chicken Curry	Pol Roti with Fish Ambul Thiyal	Red Rice with Green Gram Curry, Polos Ambula, Indian Fish Curry	778.392	1447.04	83.0972	123.12120000000000
Red Rice with Cabbage Mallung, Ash Plantain Curry, Jaffna Crab Curry	Red Rice with Green Banana Curry, Seer Fish Curry, Jackfruit Mallum	Vegetable Fried Rice with Parippu, Cashew Curry	740.595	1717.3600000000000	57.15715	152.110
Red Rice with Beetroot Curry, Capsicum Curry, Fish Curry	Seafood Fried Rice with Devilled Chicken, Devilled Fish	Red Rice with Green Gram Curry, Radish Sambol, Tropical Salad	885.3580000000000	1503.33	80.0949	92.10440000000000
Red Rice with Pineapple Curry, Bottle Gourd Curry, Fish Patties	Egg Fried Rice with Chicken Curry, Ala Badun	Egg Kottu with Devilled Chicken	596.54	1840.03	78.27665	112.68215000000000
Vegetable Fried Rice with Green Pea Curry, Fish Cutlets	Red Rice with Devilled Chicken, Murunga Curry, Kiri Hodi	Chicken Fried Rice with Chicken Curry, Ala Badun	609.425	1953.4500000000000	90.54835000000000	117.966
Fish Biryani with Ginger Chicken, Chickpea Salad	Fish Biryani with Devilled Chicken, Chickpea Salad	Red Rice with Eggplant Curry, Snake Gourd Curry, Kappa Masala	1018.6400000000000	1784.4700000000000	89.59950000000000	64.91310000000000
Red Rice with Malu Mirisata, Octopus Curry, Kappa Masala	Chicken Fried Rice with Devilled Chicken, Cashew Curry	Red Rice with Bitter Leaf Curry, Pumpkin curry, Dhal Vade	987.21	2143.29	93.41120000000000	144.682
Red Rice with Fish Ambul Thiyal, Mint Sambol, Tender Jackfruit Fry	Red Rice with Dhal Curry, Green Gram Curry, Cucumber Sambol	Red Rice with Murunga Curry, Kiri Hodi, Maalu Badun	997.479	2226.4000000000000	116.602	166.613
Pumpkin Soup	Chicken Koththu with Chicken Curry	Red Rice with Egg Curry, Collard Greens Curry, Ulundu Wade	2880.29	1364.94	67.6232	104.011
Kurakkan Roti with Malu Mirisata	Seafood Kottu with Chicken Curry	String Hoppers with Egg Curry	619.8	1217.63	66.17490000000000	45.012
Red Rice with Bitter Gourd Curry, Prawn Curry, Kiri Hodi	Masala Dosa	Red Rice with Elumas Curry, Bean Curry, Tomato Curry	328.25	1568.56	61.89230000000000	146.06280000000000
Seafood Fried Rice with Devilled Fish, Tapioca Chips	Red Rice with Sesame Seed Sambol, Collard Greens Curry, Beetroot curry	Vegetable Kottu with Devilled Fish	3410.8670000000000	1675.2500000000000	66.63440000000000	90.41650000000000
Red Rice with Fish Curry, Cashew Curry, Simmered Fish Roe	Egg Fried Rice with Devilled Fish, Fish Cutlets	Chicken Fried Rice with Devilled Fish, Ginger Chicken	1896.83	2222.81	154.52615000000000	85.7890
Red Rice with Devilled Chicken, Ginger Chicken, Mango Chutney	Chicken Fried Rice with Cashew Curry, Tapioca Chips	Red Rice with Jackfruit Curry, Bathala Curry, Mint Sambol	894.967142857	2068.09	86.8247	117.85970000000000
Red Rice with Drumstick Curry, Shrimp Cutlets, Polos Ambula	Red Rice with Bitter Gourd Salad, Thalapath Fish Curry, Ash Plantain Curry	Red Rice with Jackfruit Curry, Beetroot Curry, Carrot curry	476.95	1336.1600000000000	51.5763	135.822
Seafood Fried Rice with Devilled Chicken, Tapioca Chips	Egg Fried Rice with Devilled Fish, Tapioca Chips	Seafood Kottu with Chicken Curry	899.017	1927.58	83.46305000000000	77.1386
Vegetable Fried Rice with Cashew Curry, Vegetable Cutlets	Seafood Fried Rice with Ala Badun, Octopus Curry	Chicken Biryani with Devilled Chicken	841.442	2003.75	79.44525000000000	83.488
Egg Fried Rice with Chicken Curry, Tapioca Chips	Vegetable Soup	Red Rice with Eggplant Curry, Snake Gourd Curry, Cauliflower Pickle	424.66000000000000	1625.42	41.15375	121.1623
Red Rice with Bathala Curry, Devilled Fish, Murukku	Seafood Birvani with Devilled Beef	Chicken Birvani with Devilled Beef	1124.455	2078.472	100.27031	100.9158

- BASED ON MEALS GENERATED
- CALCULATE PRICE & NUTRITIONS ALGORITHMICALLY

MODEL BUILDING

Meal Plan Recommendation Process



The app uses fuzzy logic to map user data to a nutrient profile, calculates BMR for calorie needs, and applies a KNN model with cosine similarity to recommend personalized meal plans based on health and budget

- Step 1: User Data Collection
- Step2: Maps user data to a nutrient profile using fuzzy logic
- Step 3: BMR Calculation
- Step 4: KNN Model Training

MODEL BUILDING

STEP 1 : USER DATA COLLECTION

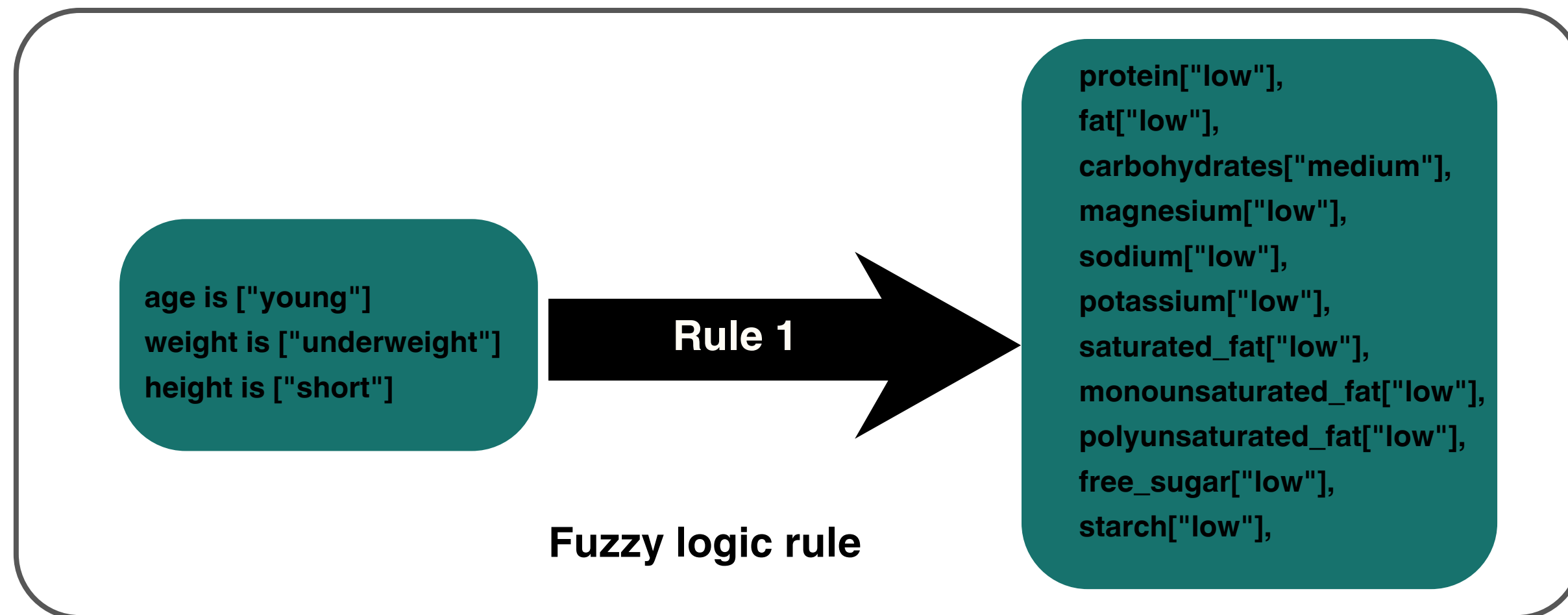


The app In survey phase collects key data such as age, health conditions, nutritional needs, Diabetic Level , Pressure level , Cholesterol level and budget to begin the meal plan customization process.

MODEL BUILDING

STEP 2 : MAPS USER DATA TO A NUTRIENT PROFILE USING FUZZY LOGIC

In the fuzzy logic phase, takes user input data like age, weight, health conditions, and dietary preferences, diseases and using fuzzy logic sets maps them to a optimal user's nutritional profile .



MODEL BUILDING

- Gradual Membership:

Fuzzy logic allows for varying degrees of membership in categories (e.g., "medium protein," "low carbs") rather than strict binary classifications

- Handling Uncertainties:

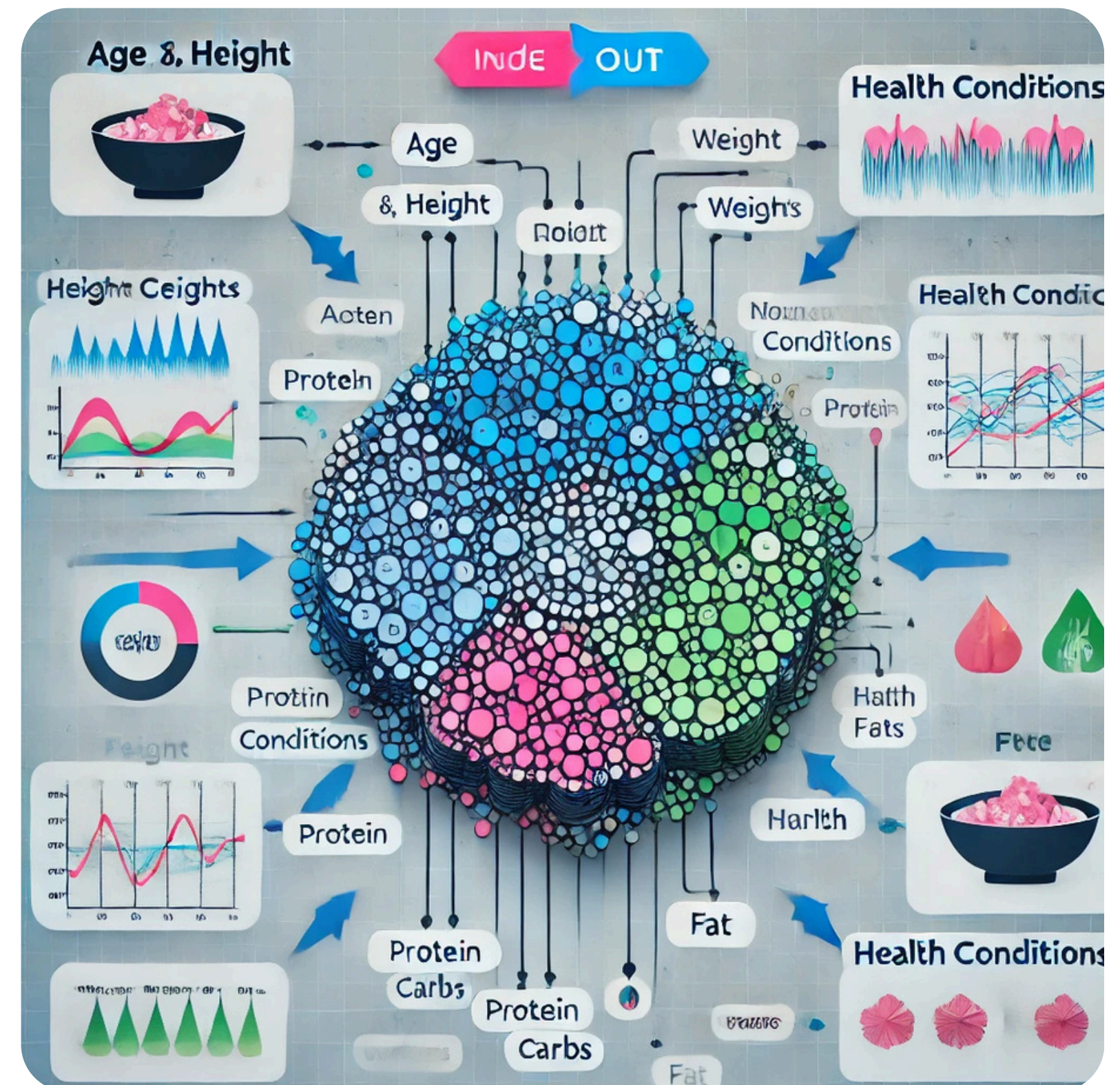
This approach effectively addresses the uncertainties and variability in individual health and nutritional needs, acknowledging that requirements can differ widely.

- Creation of Fuzzy Sets:

For each nutritional factor (such as calories, protein, fats), fuzzy sets are created to represent different levels (e.g., low, medium, high) rather than fixed thresholds.

- Overlapping Rules:

Overlapping fuzzy rules are established, which means that multiple nutritional factors can be considered simultaneously to determine the ideal composition.



MODEL BUILDING

STEP 3 : BMR CALCULATION

To calculate BMR, gather user data: age, gender, height, and weight. Use the appropriate formula for men or women, substituting the values, then perform the calculation to determine the Basal Metabolic Rate.



Daily calories are determined by multiplying BMR by an activity factor based on the user's activity level.

BMR Formula

(Mifflin - St Jeor)



MEN

BMR =
 $(4.536 \times \text{weight in lbs})$
 $+ (15.88 \times \text{height in inches})$
 $- (5 \times \text{age}) + 5$

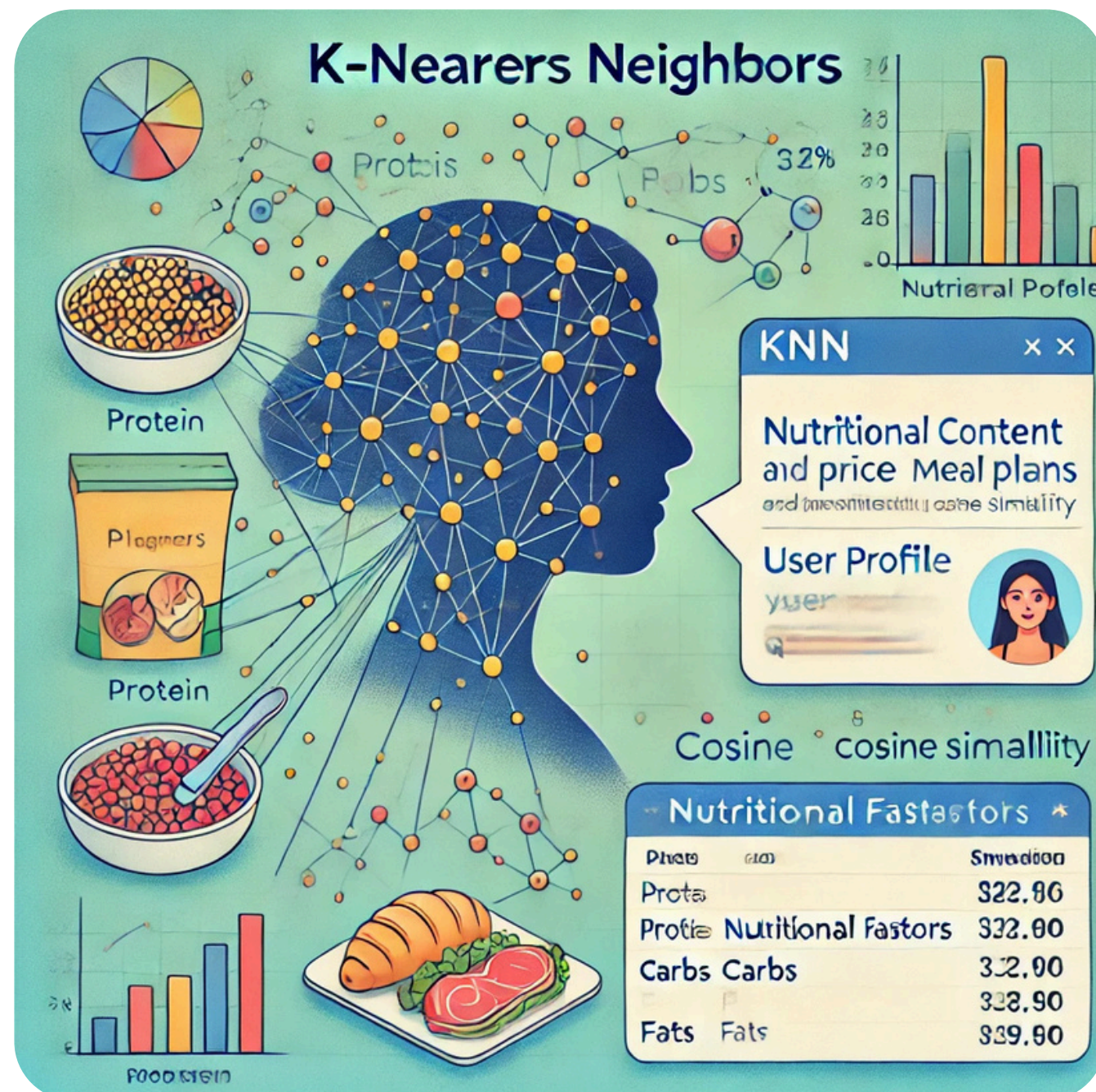


WOMEN

BMR =
 $(4.536 \times \text{weight in lbs})$
 $+ (15.88 \times \text{height in inches})$
 $- (5 \times \text{age}) - 161$

MODEL BUILDING

STEP 4: KNN MODEL TRAINNING



The K-Nearest Neighbors (KNN) model is trained using nutritional content and meal prices. Each meal plan is represented as a data point with its nutritional attributes. When a user submits their profile, map user data to their nutrient profile using Fuzzy Logic and then cosine similarity is computed between their profile and available meal plans, identifying the closest matches based on their nutritional requirements, health conditions, and budget.

ChatBot Integration

- **CoolBot:** Enhances user experience with detailed food information and Meal Recipies.
- **RAG Architecture :** use RAG architecture to build customized chat bot
- **Gemini API:** Provides accurate insights on ingredients, nutrition, and meal options.

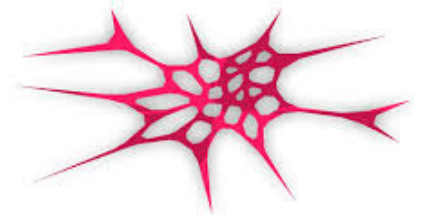


LangChain



Gemini

FAISS
Scalable Search With Facebook AI

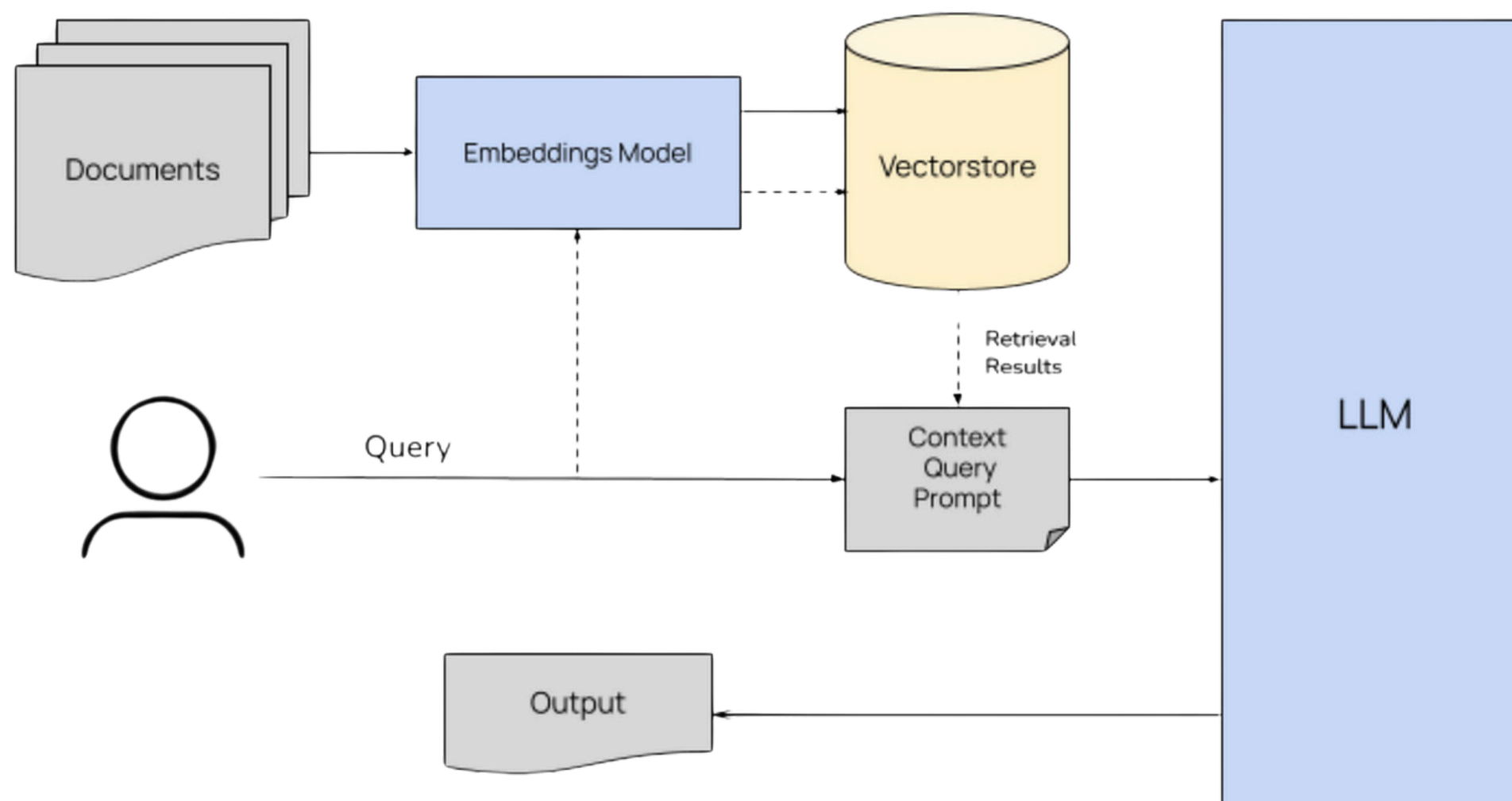


RAG

Architecture

Data Source :

- PDFs: Extracts food-related data from PDF documents.
- Embedding Process: Converts PDF content into vector format using Google embeddings.



- **Langchain:** Framework for building the chatbot architecture.
- **Gemini-pro:** Large Language Model (LLM) for generating intelligent responses.
- **Google Embeddings:** Converts food-related text data to vectors for similarity search.
- **Faiss (Facebook AI Similarity Search):** Fast vector database for efficient data retrieval.
- **MongoDB:** Manages user chat histories for personalized conversations.

MODEL EVALUATION



- Conducted a Google Form survey among friends and sample users to evaluate the Sri Lankan Meal Recommendation System.
- Assess how well recommendations align with local dietary preferences.
- Key data collected included user satisfaction, dietary preferences, and nutritional adequacy.
- Analyzed feedback to measure model accuracy and identify areas for improvement.

GOOGLE FORM SURVEY

Feedback on Accuracy of CoolMeal

Hello! We're gathering feedback on the **accuracy** of our meal plan recommendation model.

First, check out the app here: <https://coolmeal.vercel.app/>,

then please take a moment to share your thoughts by filling out this form. Your input is invaluable to us. Thank you!

lakshan.21@cse.mrt.ac.lk

Switch accounts

Not shared

* Indicates required question

Get a recommendation from here: <https://coolmeal.vercel.app/>



COOLMEAL

1. On a scale of 1-5, how accurately did the recommended meal plan align with your dietary needs and preferences?

*

1. On a scale of 1-5, how accurately did the recommended meal plan align with your dietary needs and preferences?

15 responses

[Copy chart](#)

Average rating (4.07)



Rating	Count	Percentage
1	1	6.7%
2	1	6.7%
3	2	13.3%
4	3	20%
5	8	53.3%

The overall meal plan seems to be good. It's great showing the price of each recommendations so the users know how much they are gonna spend on their meals

Align with the daily price for meals accurately, but applicability of meals is not very effective.

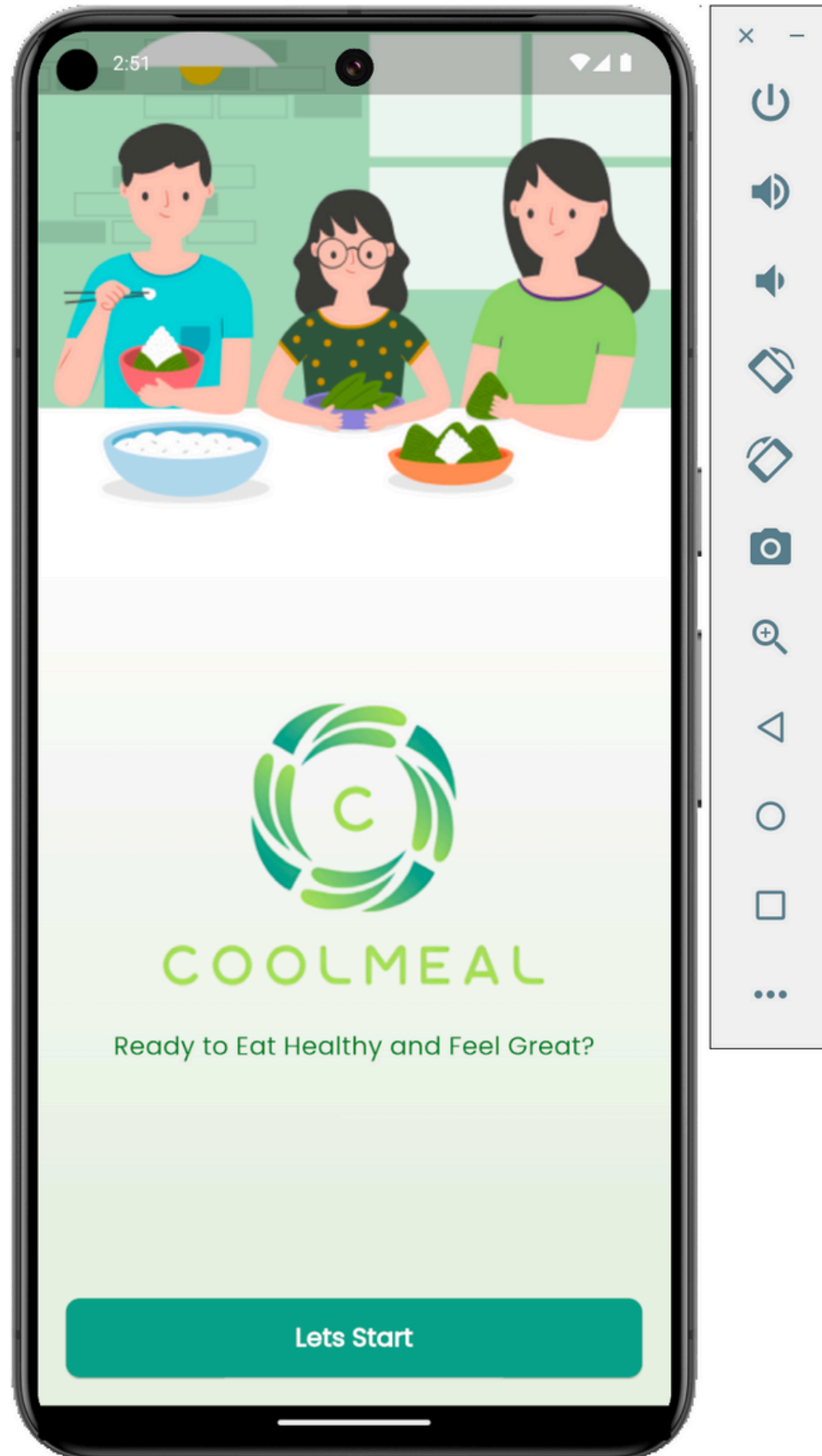
health concerns

satisfied

MODEL EVALUATION

CHATBOT EVALUATION

- Evaluation Methodology
 - Measured chatbot responses by comparing answers with data directly retrieved from PDF files.
 - Accuracy metrics focused on response relevancy and factual correctness.
- Key Metrics & Analysis
 - Accuracy: Percentage of responses correctly matched with PDF data.
 - Relevancy: Ensured answers were contextually aligned with user queries and supported by PDF content.



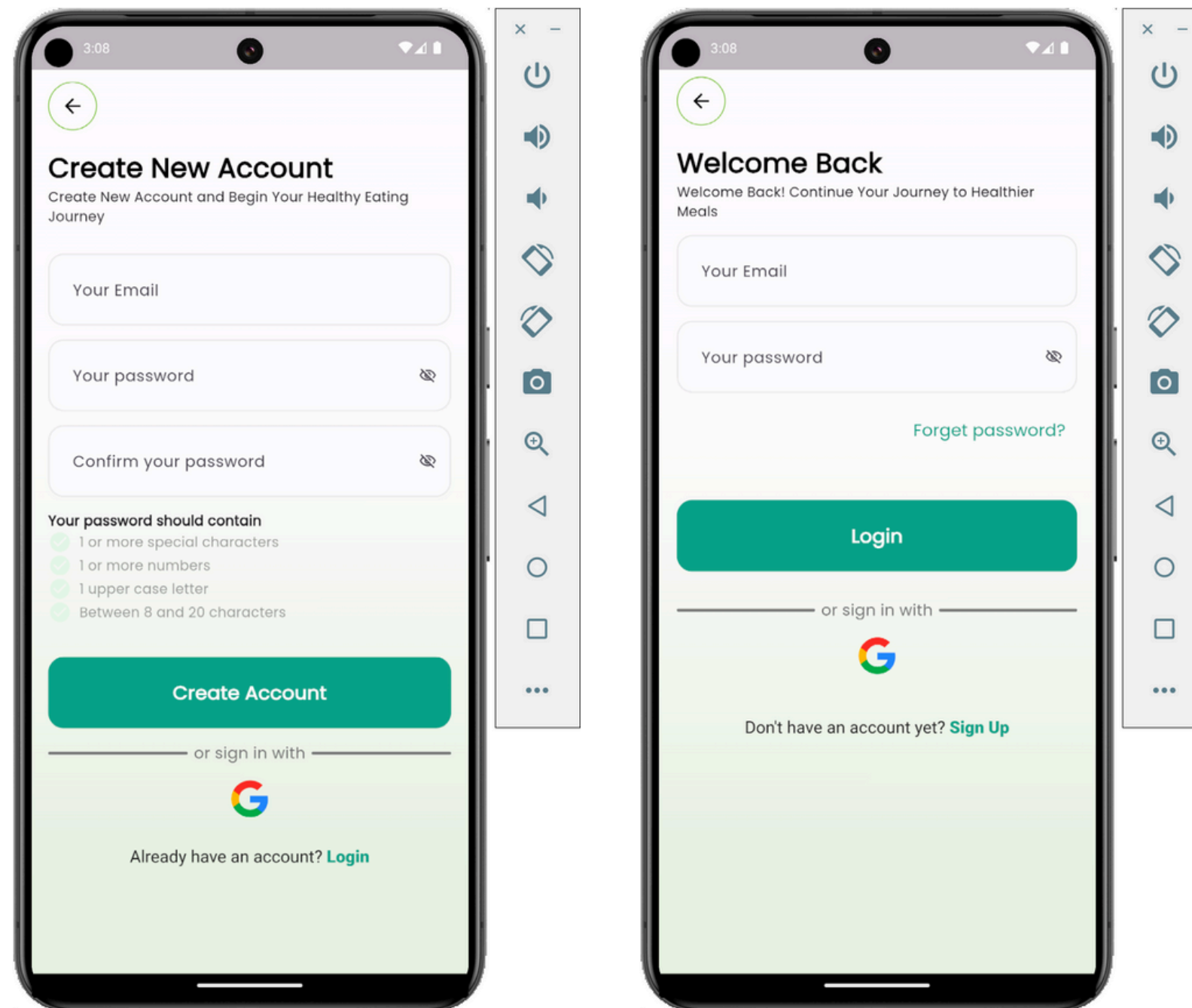
COOLMEAL

MOBILE APP

COOLMEAL APP

A user-friendly mobile platform that provides personalized meal plans, real-time suggestions, and easy interaction with an AI-powered chatbot.

MOBILE APP FEATURES |



AUTHENTICATION

COOLMEAL USES FIREBASE AUTH FOR SECURE AND FLEXIBLE USER AUTHENTICATION, SUPPORTING MULTIPLE SIGN-IN METHODS AND SEAMLESS INTEGRATION WITH OUR FLUTTER APP.

MOBILE APP FEATURES |

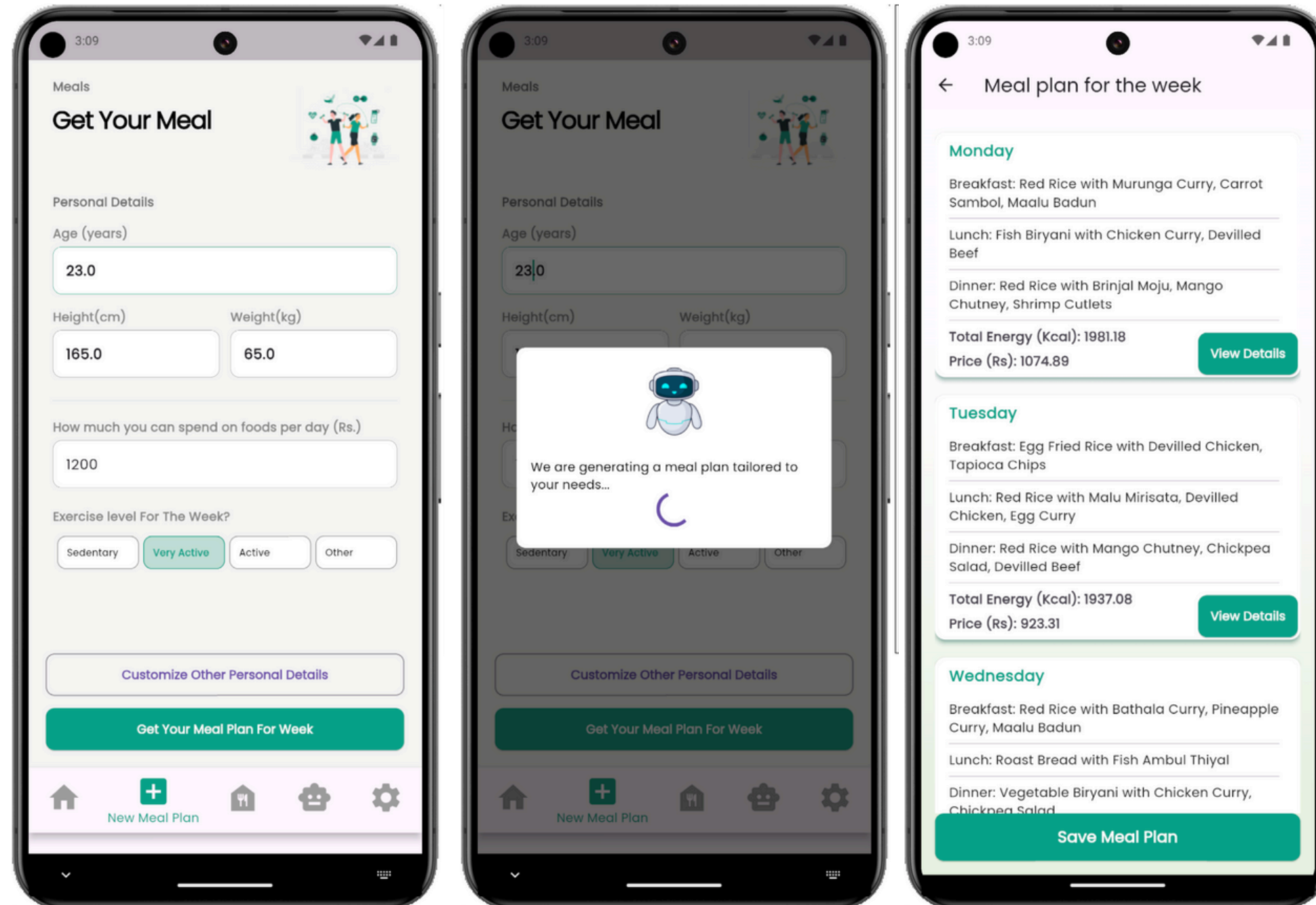
The image displays three sequential screens of a mobile application for completing a user profile. Each screen has a status bar at the top showing the time as 3:09 and standard mobile connectivity icons. A 'Skip' link is located in the top right corner of each screen.

- Screen 1: Complete Your Profile**
 - Header: Complete Your Profile
 - Illustration: A group of four people (two men and two women) in athletic wear performing various exercises.
 - Section: About You
 - Text: Tell Us About Yourself to Personalize Your Meal Plans.
 - Icon: A person with a checkmark.
 - Section: How to connect with you
 - Text: Share Your Health & Fitness Details for Customized Plans.
 - Icon: A group of people.
 - Button: Get Started
- Screen 2: Tell more about yourself**
 - Header: Tell more about yourself
 - Form fields:
 - Name: Pinsara Mallawarachchi
 - Height(cm): 165
 - Weight(kg): 65
 - Age(years): 12
 - Gender: Male (selected), Female
 - Button: Next
- Screen 3: Tell Me More About You**
 - Header: Tell Me More About You
 - Form fields:
 - Diabetes Level: None (selected), Medium, High, Very High
 - Pressure Level: None, Medium (selected), High, Very High
 - Cholesterol Level: None, Medium (selected), High, Very High
 - Any allergies?: Beef
 - Fitness goals?: Cutting, Bulking, Weightloss (selected)
 - Exercise level?: Sedentary, Very Active (selected), Active, Other
 - Button: Next

USER SURVEY ON INITIAL LOGIN

COOLMEAL COLLECTS USER DATA SUCH AS AGE, WEIGHT, HEIGHT, HEALTH CONCERNS, AND ALLERGIES TO PROVIDE PERSONALIZED MEAL PLAN RECOMMENDATIONS.

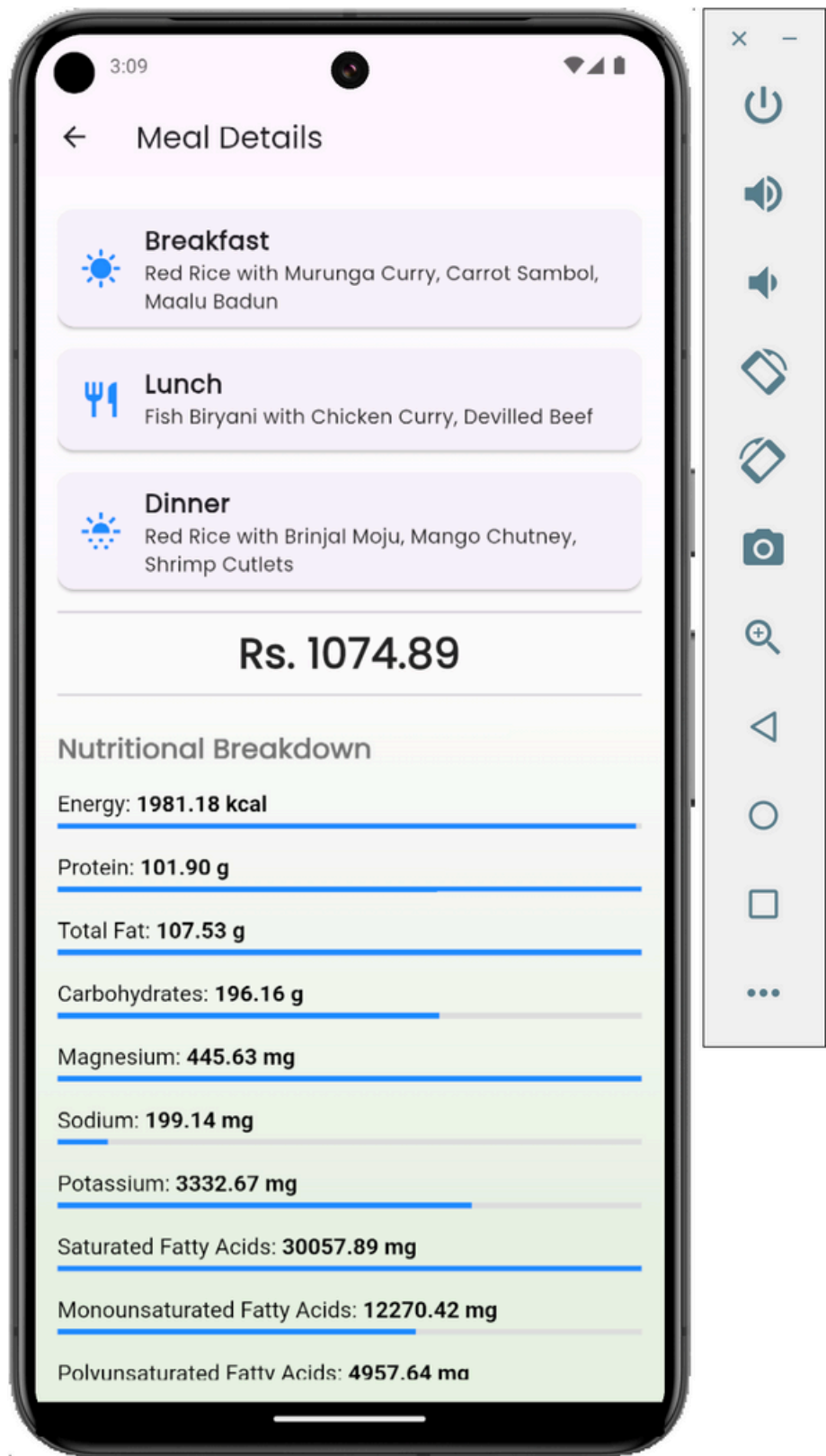
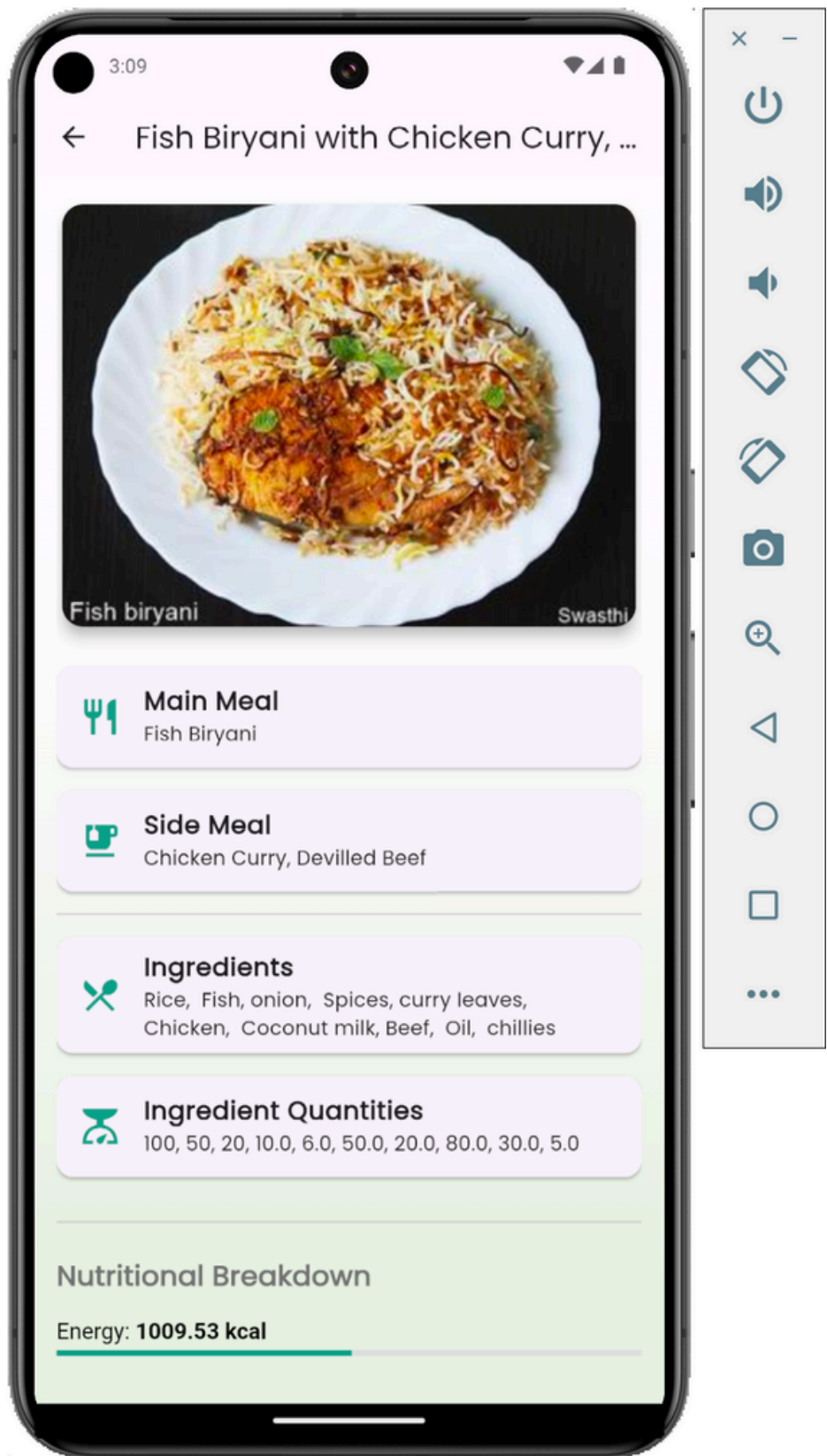
MOBILE APP FEATURES |



**GENERATE MEAL PLAN FOR
UPCOMING WEEK**

**COOLMEAL GENERATES THE MOST SUITABLE MEAL
PLAN FOR THE UPCOMING WEEK, TAILORED TO
USERS' NUTRITIONAL NEEDS AND PREFERENCES**

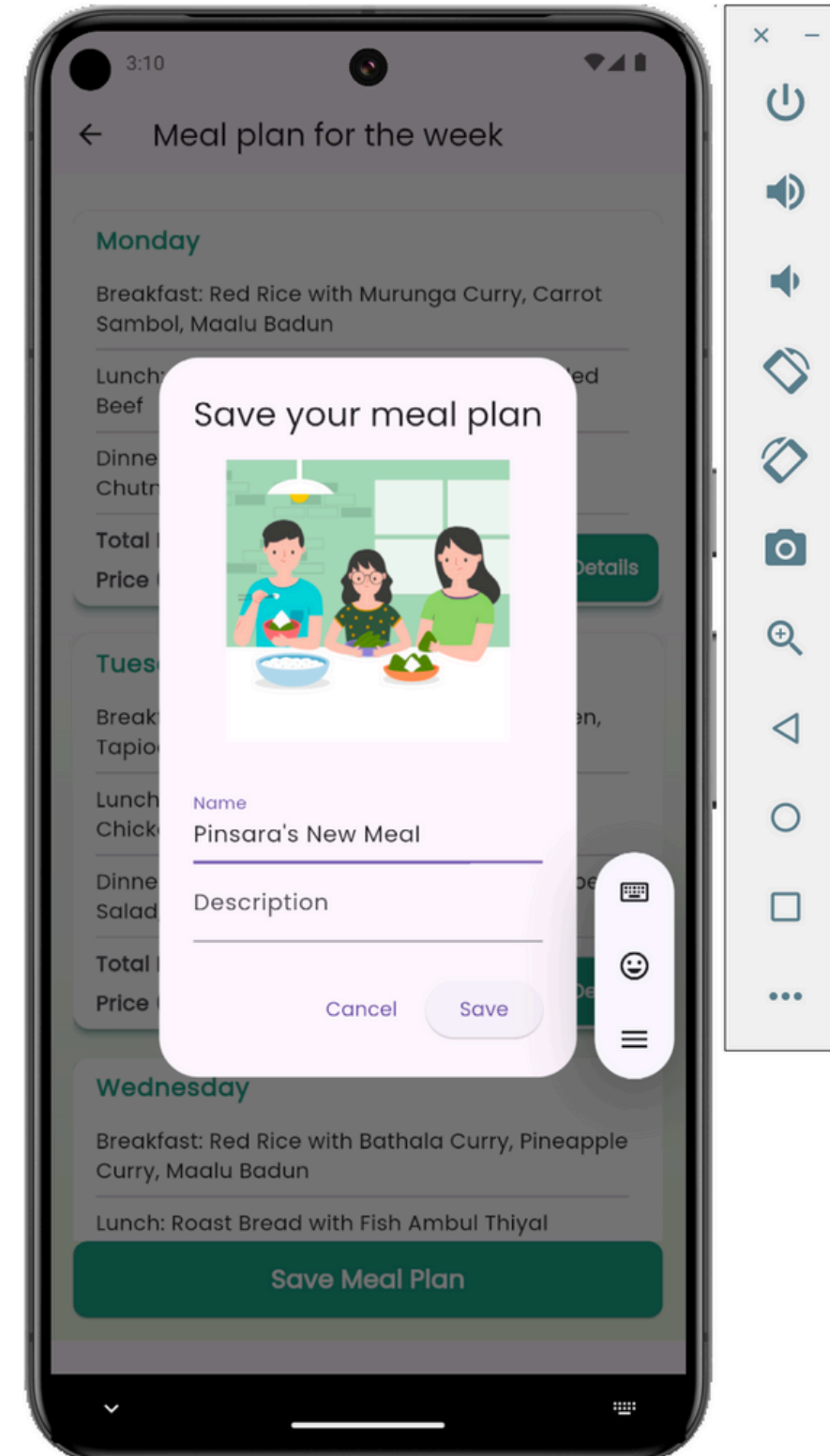
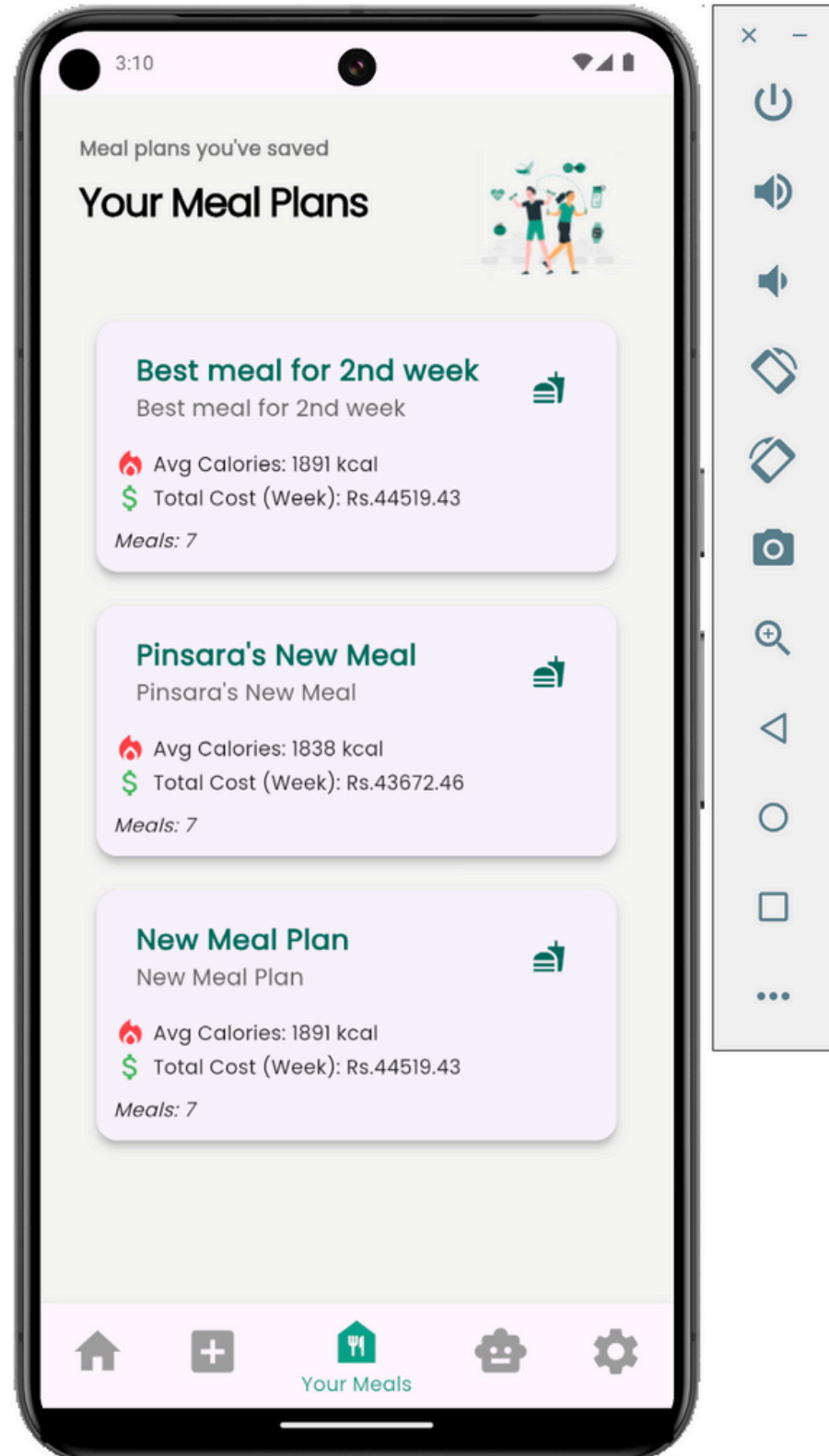
MOBILE APP FEATURES |



NUTRITIONAL BREAKDOWN

COOLMEAL ENABLES USERS TO VIEW DETAILED NUTRITIONAL CONTENT AND PRICES FOR BOTH MEAL PLANS AND INDIVIDUAL MEALS, HELPING THEM MAKE INFORMED CHOICES.

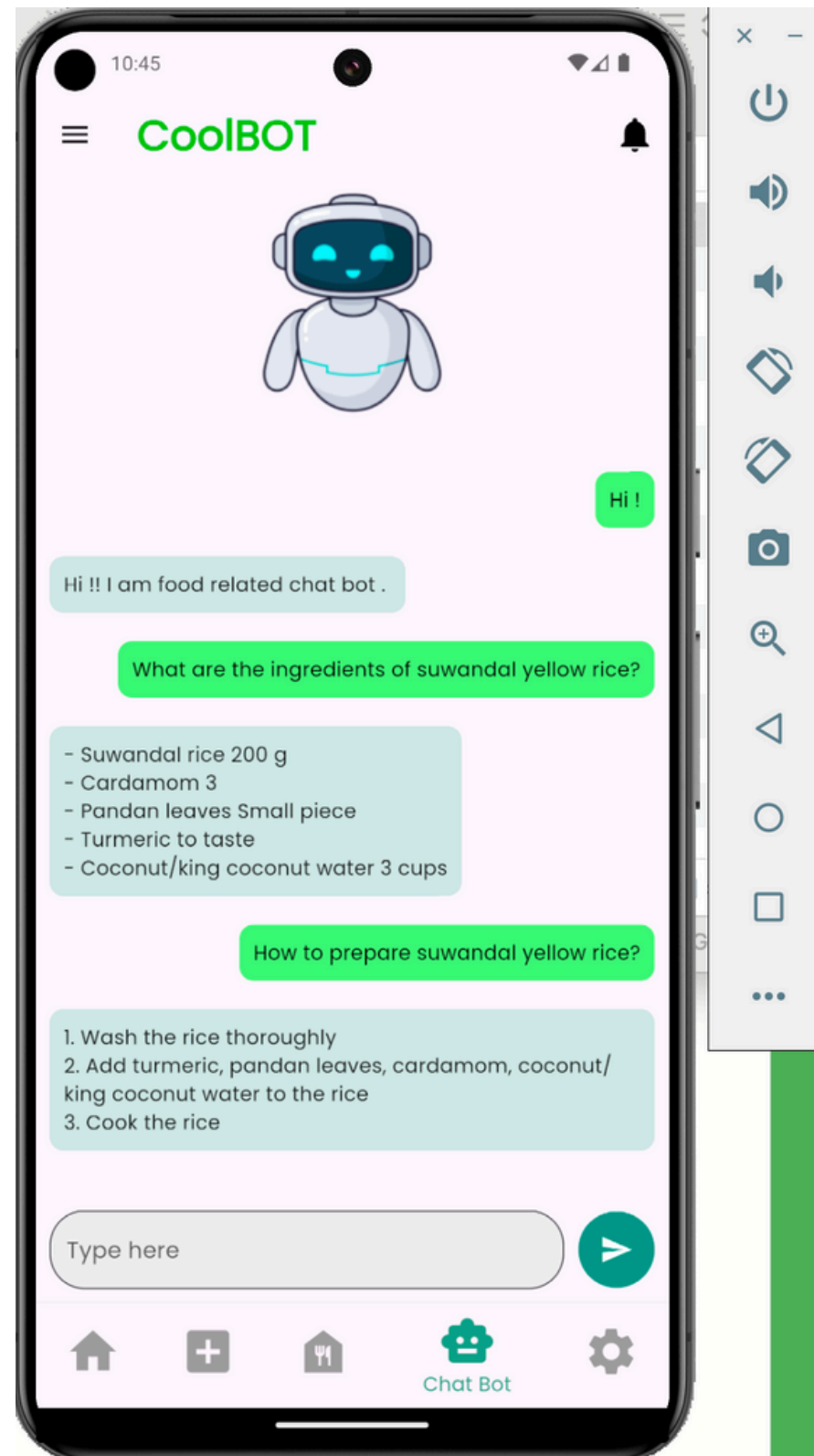
MOBILE APP FEATURES |



SAVE GENERATED MEALS

COOLMEAL ALLOWS USERS TO SAVE GENERATED MEAL PLANS FOR FUTURE REFERENCE, ENSURING EASY ACCESS TO PERSONALIZED RECOMMENDATIONS

MOBILE APP FEATURES |

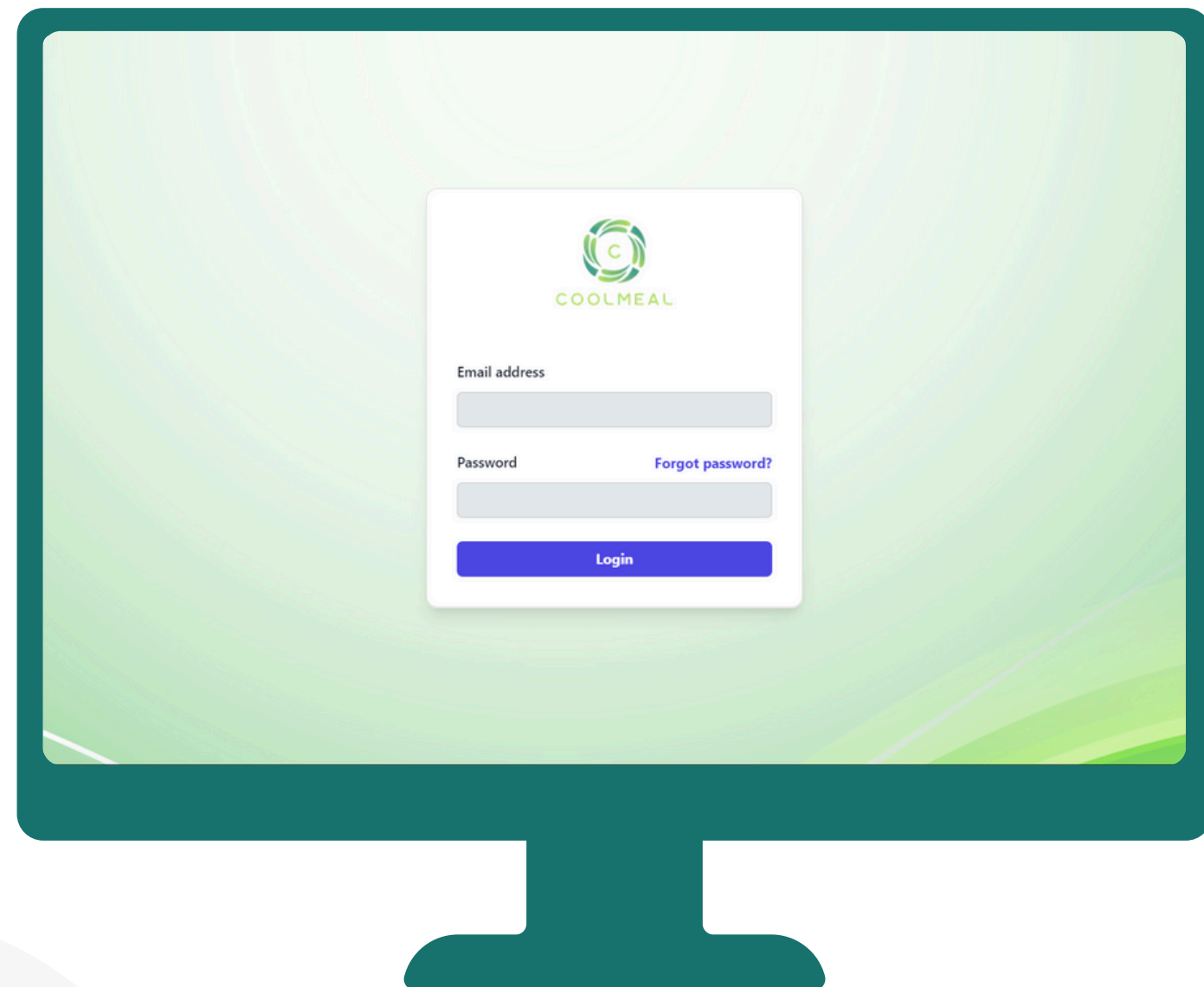


COOLBOT

COOLMEAL ALLOWS USERS TO SAVE GENERATED MEAL PLANS FOR FUTURE REFERENCE, ENSURING EASY ACCESS TO PERSONALIZED RECOMMENDATIONS



COOLMEAL



ADMIN PANEL

COOLMEAL ADMIN

A comprehensive control interface for administrators to manage user data, customize meal plans, Add data to customized chat bot and monitor system performance.



COOLMEAL

Admin Panel : Add New Meal Plan

Cool Admin

Manage

Dashboard

Users

Food Ingredients

Meal Items

Meals

Meal Plans

Chatbot

Analytics

Model

Trend

Add New Meal

Submit a new meal for inclusion in the meal planning database.

Breakfast

Main Meal

Enter breakfast main meal

Side Meal

Enter breakfast side meal

Complete Meal

Enter breakfast complete meal

Breakfast Probability

Enter breakfast probability

Lunch Probability

Enter lunch probability

Dinner Probability

Enter dinner probability

Combined Meal

Enter combined meal count

Ingredients

Enter ingredients

Quantities

Enter quantities

Energy (Kcal)

Enter energy value

Protein (g)

Enter protein amount

Total Fat (g)

Enter total fat amount

Carbohydrates (g)

Total Dietary Fibre (g)

Vitamin A (µg)

Search

Logout

TOTAL FAT (G)

83.1566

TOTAL CARB

175

116.79305

113

This admin panel allows for adding detailed meal information, including main and side meals, nutritional values (calories, protein, fat), ingredients, and meal probabilities. Once added, the meal plans are used to retrain the recommendation model, ensuring personalized and accurate meal suggestions for users.



COOLMEAL

Chatbot Training : PDF Upload Page

This page allows the admin to upload PDF files containing food-related data. These files are used to retrain the chatbot, improving its accuracy and relevance for meal recommendations and user interactions.

Manage

Dashboard

Users

Food Ingredients

Meal Items

Meals

Meal Plans

Chatbot

Analytics

Model

Trend

Upload PDF files to Train Food Related Chatbot

Upload training data in PDF format to improve chatbot accuracy.

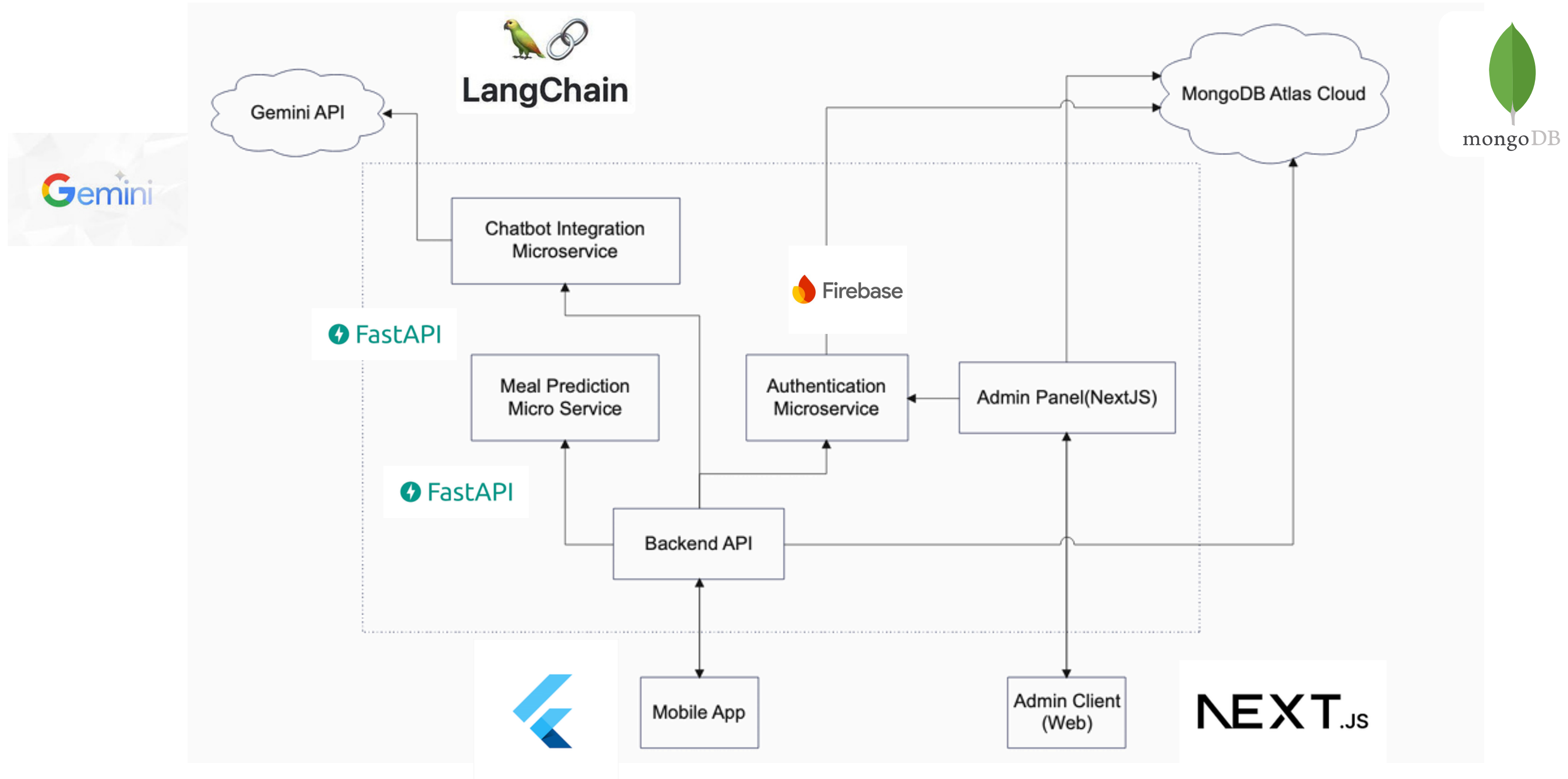
Upload PDF

Select a PDF file:

Choose FileNo file chosen

Upload PDF

SYSTEM ARCHITECTURE



BACKEND TESTING

Unit testing uses Pytest for isolated function testing on both servers. GitHub Actions automates tests on updates, ensuring smooth CI/CD by verifying successful deployments.

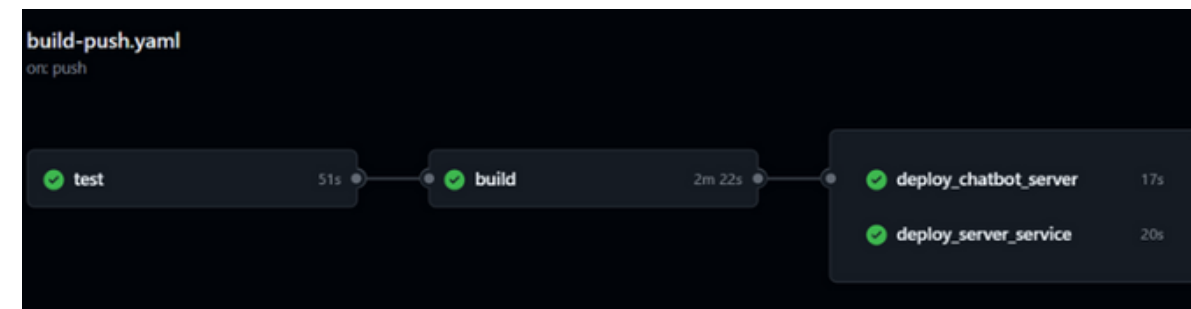
```
(venv) (base) PS D:\DSE project\coolmeal\server> pytest test_main.py -v --tb=long --durations=10 --junitxml=report.xml
D:\DSE project\coolmeal\server\venv\lib\site-packages\pytest_asyncio\plugin.py:208: PytestDeprecationWarning: The configuration option "asyncio_default_fixture_loop_scope" is unset.
The event loop scope for asynchronous fixtures will default to the fixture caching scope. Future versions of pytest-asyncio will default the loop scope for asynchronous fixtures to function scope. Set the default fixture loop scope explicitly in order to avoid unexpected behavior in the future. Valid fixture loop scopes are: "function", "class", "module", "package", "session"

  warnings.warn(PytestDeprecationWarning(_DEFAULT_FIXTURE_LOOP_SCOPE_UNSET))

===== test session starts =====
platform win32 -- Python 3.11.7, pytest-8.3.3, pluggy-1.5.0 -- D:\DSE project\coolmeal\server\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\DSE project\coolmeal\server
plugins: anyio-4.4.0, asyncio-0.24.0
asyncio: mode=Mode.STRICT, default_loop_scope=None
collected 5 items

test_main.py::test_check_root PASSED [ 20%]
test_main.py::test_read_prediction PASSED [ 40%]
test_main.py::test_read_prediction_check_all_attr PASSED [ 60%]
test_main.py::test_missing_input_data PASSED [ 80%]
test_main.py::test_invalid_input_data PASSED [100%]

----- generated xml file: D:\DSE project\coolmeal\server\report.xml -----
slowest 10 durations:
1.68s call     test_main.py::test_read_prediction
1.63s call     test_main.py::test_read_prediction_check_all_attr
0.01s call     test_main.py::test_check_root
(7 durations < 0.005s hidden. Use -vv to show these durations.)
===== 5 passed in 5.38s =====
(venv) (base) PS D:\DSE project\coolmeal\server>
```



GitHub Actions

```
(venv) (base) PS D:\DSE project\coolmeal\chatbot-service\server> pytest test_main.py -v
===== test session starts =====
platform win32 -- Python 3.11.7, pytest-8.3.3, pluggy-1.5.0 -- D:\DSE project\coolmeal\chatbot-service\server\myenv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\DSE project\coolmeal\chatbot-service\server
plugins: anyio-4.4.0
collected 6 items

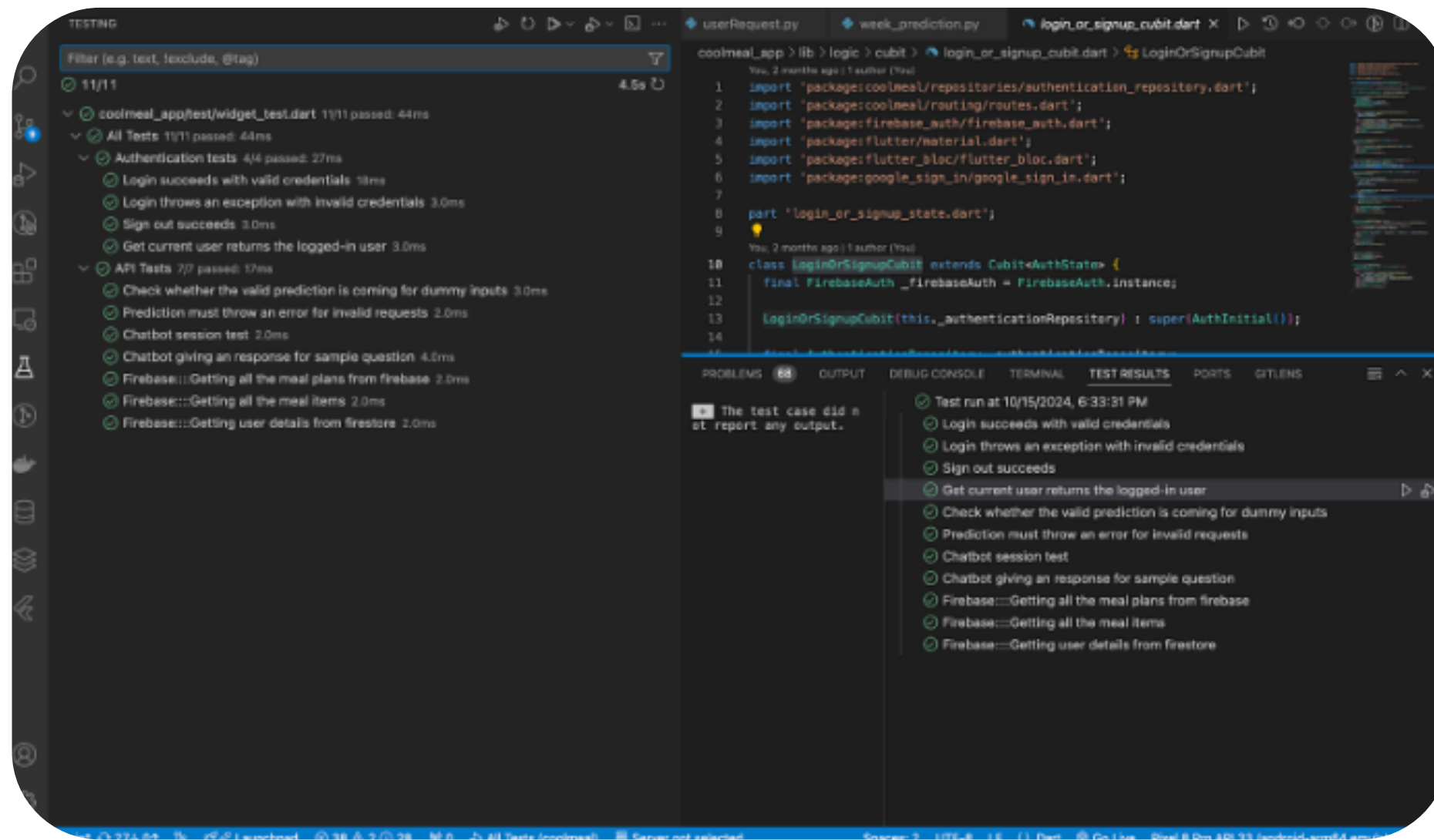
test_main.py::test_check_root PASSED [ 16%]
test_main.py::test_chat PASSED [ 33%]
test_main.py::test_get_session PASSED [ 50%]
test_main.py::test_insert_session_data PASSED [ 66%]
test_main.py::test_insert_session_data_not_found PASSED [ 83%]
test_main.py::test_get_history PASSED [100%]

----- warnings summary -----
test_main.py::test_get_history
D:\DSE project\coolmeal\chatbot-service\server\createContext.py:98: LangChainDeprecationWarning: The method `BaseLLM.__call__` was deprecated in langchain-core 0.1.7 and will be removed in 1.0. Use invoke instead.
  summary = llm(summary_prompt)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 6 passed, 1 warning in 102.74s (0:01:42) =====
(myenv) (base) PS D:\DSE project\coolmeal\chatbot-service\server>
```

MOBILE APP TESTING

- employ the Mockito library for testing, allowing efficient unit tests through mock objects to simulate dependencies.

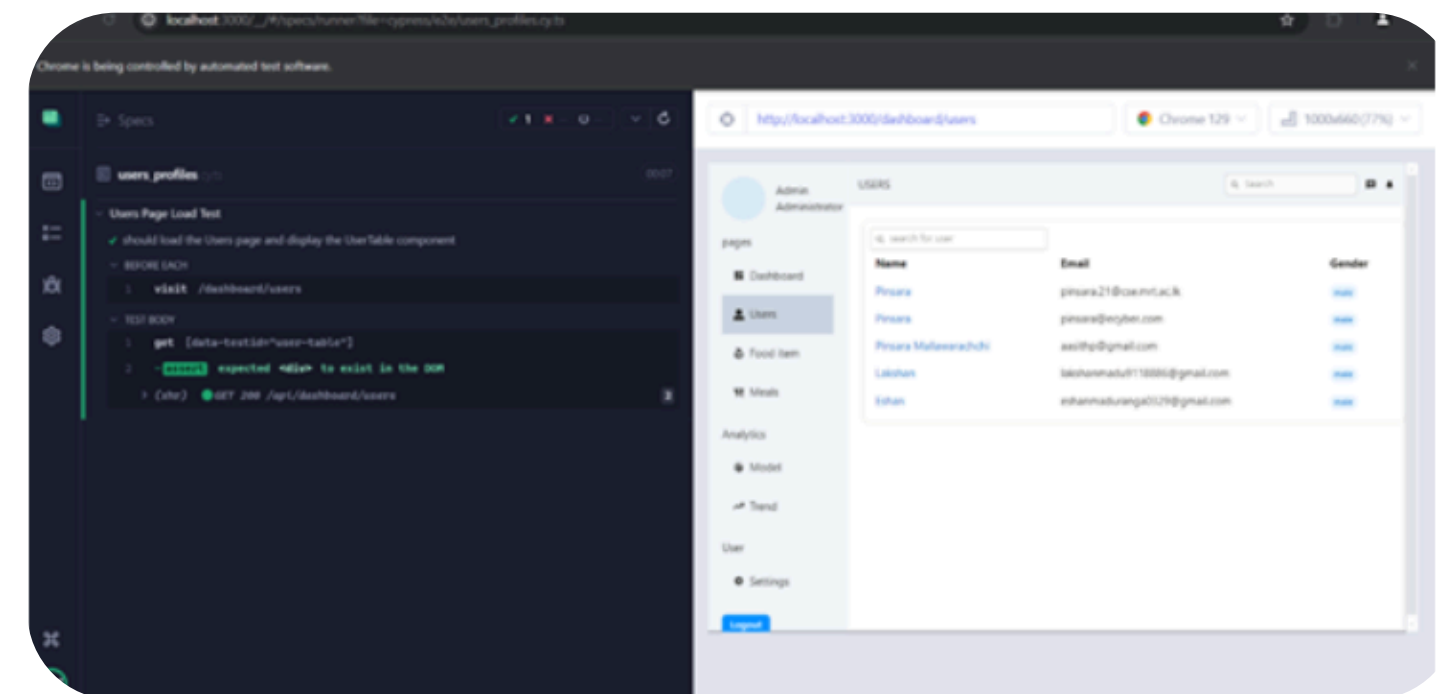
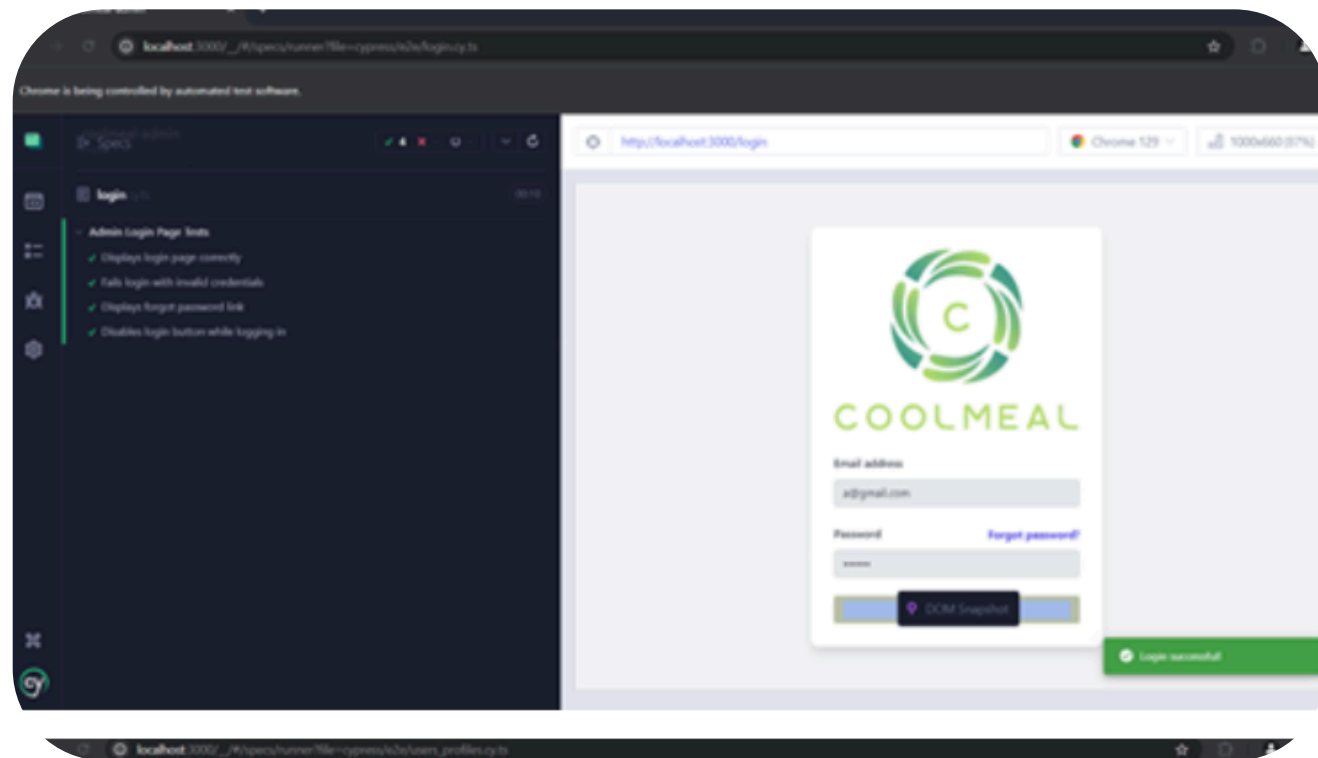


mockito



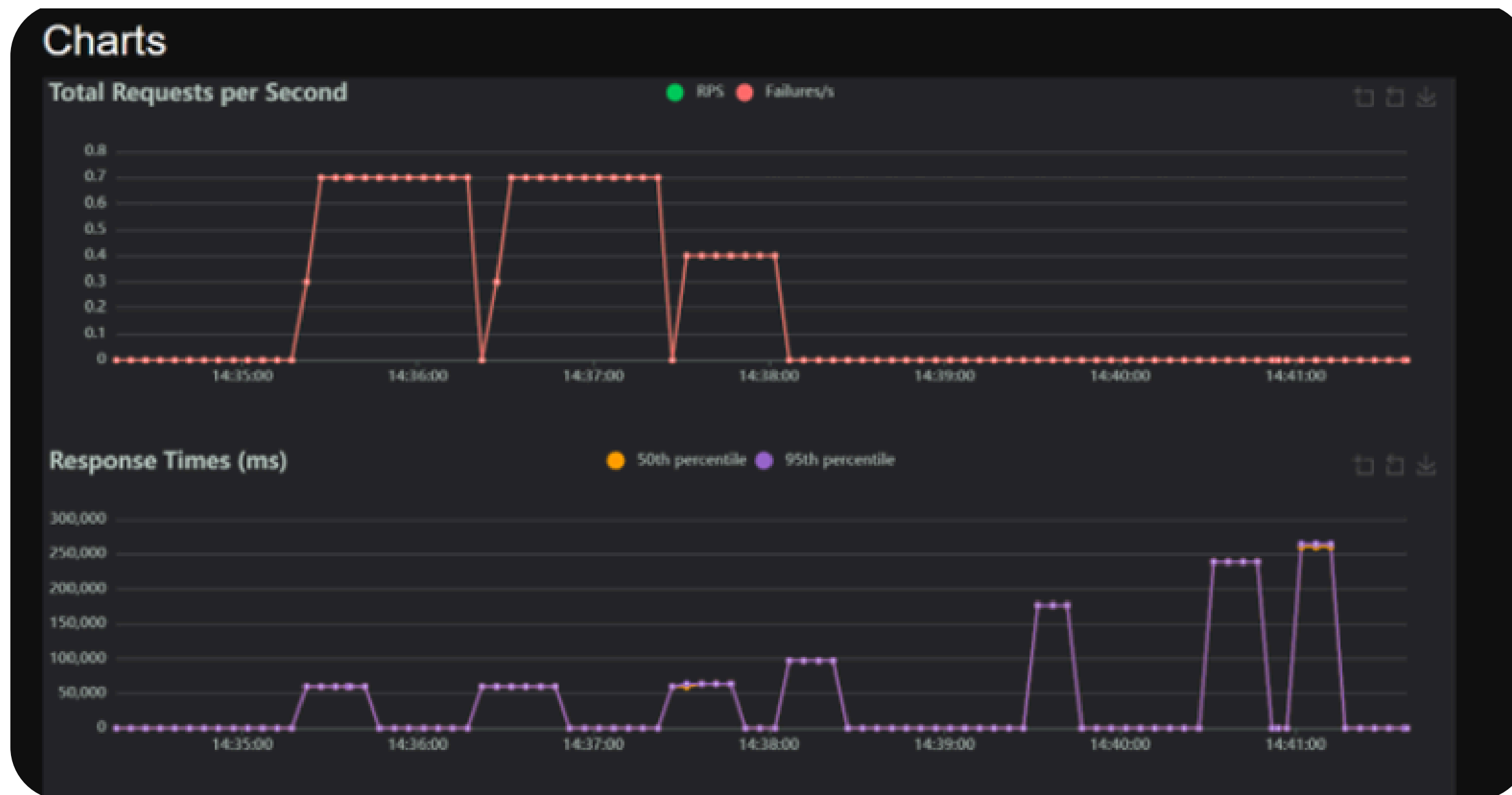
WEB UI TESTING

- For the admin panel, we use **Cypress** to test the web UI built with Next.js, enabling end-to-end testing for a reliable user interface experience.



LOAD TESTING

- For load testing, we used Locust, an open-source tool, to simulate user traffic and assess server performance. Locust allows testing under different load conditions, ensuring application resilience and scalability.



LOCUST

Use of Technology



Mobile App
Flutter



Admin panel
Next.js
Tailwind CSS
Radix UI



Backend
FastAPI



Deployment
AWS
Docker
Github actions



ML
Scikitlearn
Fuzzy logic
Pandas
Langchain
FAISS
Gemini API

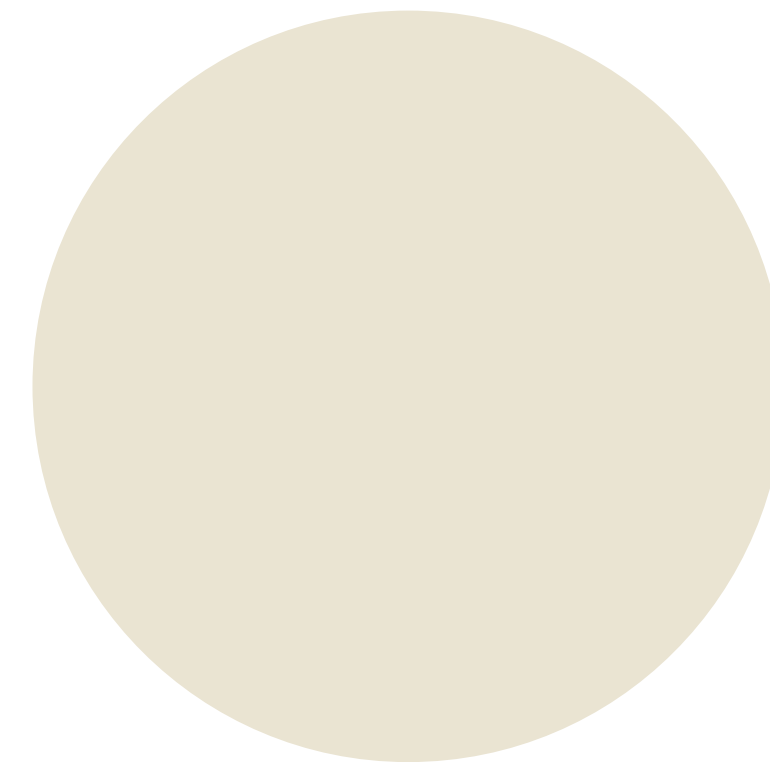


Database
Mongodb
Firebase



THANK YOU

● FOR YOUR NICE ATTENTION



DEMO

I AN INTERACTIVE PREVIEW HIGHLIGHTING
CORE APP FEATURES AND USER
EXPERIENCE.

